

# Supplementary Material: More Than the Risk: Frontier LLM Behaviour in AI Development Races Depends on the Rival

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## Abstract

This supplement supports More Than the Risk: Frontier LLM Behaviour in AI Development Races Depends on the Rival. It records the quality-control screen, the theory comparison, the scripted-rival evidence, and the human diversity analysis, together with prompt and game robustness checks. These materials use the same routes, task and evidence boundaries as the main paper and do not extend its claims beyond the idealised race.

## Keywords

Large language models, AI development races, multi-agent systems, repeated games, AI safety, opponent-conditioned behaviour

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## Supplementary roadmap

The supplement is organised as a short evidence package rather than a lab notebook. The unnumbered dividers identify the reader's path; the subsection labels remain attached to their original analyses so that existing cross-references stay stable.

For the headline claims, read Sections A–C first. Section D supplies the cluster-aware human comparison and its diagnostics; Section E is exploratory. The scripted-rival block is physically placed in Section C so its complete evidence follows the main paper's rival result directly.

Four terms are used with fixed meanings throughout. A *route* is the exact provider string a request is addressed to, for example `google/gemini-3-flash-preview`. An *endpoint* is a route that carries an admission verdict from the probe battery below. A *checkpoint* is one dated model version, which is what a comparison across the rows of a behavioural table compares. *Model* is used only generically, for a language model as a kind of system. Where a number belongs to a route rather than to a model in general, the route string is printed beside it.

## A Extended results and robustness checks

This supplement collects the material referenced from the main paper as supporting rather than extending the main claims: the record of the quality-control screen and which routes it admitted

(§A.1), the human-benchmark replication check (§A.8), the persona comparison (§A.11), the predictive analysis (§A.14), and the full two-player and *N*-player descriptive results (§A.15, §A.16). It also carries the evidence the main paper only summarises: the separate and larger battery on the one model with published weights (§A.2), the selection-strength sweep of the evolutionary benchmark (§A.6), the crossed representation-robustness design completed on both routes selected for the crossed representation study (§A.7), the matched group-size comparison (§A.24), the scripted-rival campaign (§A.10), the cluster-count sweep behind the human archetypes (§A.21), the trajectory-diversity statistics (§A.22), the pilot roster the projections are built on (§A.12), the joint outcome dynamics of a race (§A.25), and the designed horizon boundary at round five (§A.26).

*A. Validity, admission, and provenance.*

### A.1 The quality-control screen: what it asked, and which routes passed

This subsection records the screen as an entry condition and nothing more: what it asked, when its rule was fixed, which routes cleared it, and the three facts about it that bound what a verdict may be used for. It is deliberately short. Nothing in this supplement ranks routes by their probe scores or treats the verdict as a measurement of capability.

The screen is a frozen bank of 20 probes over six domains, asked three times each, giving 60 scored answers per route and 540 in all. The six domains are recalling the rules, reading the stage-payoff matrix, rebuilding the current state from a short history, taking one state transition, scoring a finished race, and computing a closed-form expected payoff. A route is admitted only if it clears all three of the conditions recorded in the caption of Table 2: 80% of all its answers, 75% within state reconstruction and 75% within terminal scoring. Expected payoff is recorded and never blocks admission, because the game asks an agent to choose an action rather than to solve for an expected value.

Two properties of the rule are what let it serve as an entry condition. The bank, the three conditions and their values were fixed and recorded before the campaign ran, in the written collection protocol. And the screen scores probe answers against ground truth without ever reading a race, so no gameplay result can reach a verdict, whatever order things were run in. We do not call the screen pre-registered, because three of the nine routes received their first verdict after their own gameplay had completed; what is checkable, and what the defence rests on, is the fixed bank, the fixed rule and the fact that the instrument cannot see the outcome it selects on.

Three facts make the nine rows of Table 2 comparable with one another. All nine ran under one protocol identifier. All nine answered the same probe bank, whose SHA-256 hash begins 2953fb47, and read the same rules context, whose hash begins b80981a5; the



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**Table 1: Roadmap to the supplementary evidence. Sections A–C support the headline claims most directly; Sections D–E contain the human diagnostics and exploratory extensions. The main paper’s claims remain scoped to the designs and populations stated there.**

| Section                             | Supporting analyses                                                                                                                                        | What the section adds                                                         |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| A. Validity and provenance          | Quality-control screen (§A.1); open-weight battery (§A.2); human-source reconstruction (§A.8)                                                              | Entry conditions, source checks and the boundaries of the evidence.           |
| B. Core risk, theory and robustness | Baseline and policy shape (§A.3, §A.4); replication (§A.5); selection-strength sweep (§A.6); crossed representation design (§A.7)                          | The risk response, evolutionary comparison and representation checks.         |
| C. Scripted-rival intervention      | Full five-route campaign (§A.10)                                                                                                                           | The complete fixed-policy intervention, replay audit and paired effects.      |
| D. Human comparison and diversity   | Two-player scope (§A.15); diversity (§A.22); population identity (§A.13); predictive structure (§A.14); archetype and embedding diagnostics (§A.23, §A.17) | Sequence-level comparison and diagnostics of observable trajectory diversity. |
| E. Exploratory interaction scope    | Personas (§A.11); group size (§A.16, §A.24); position, dynamics and horizon (§A.20, §A.25, §A.26); citation record (§A.9)                                  | Scoped extensions and the remaining reproducibility record.                   |

analysis script fails closed and refuses to tabulate a route whose hashes differ. Eight routes ran at benchmark task version 7 and one, `google/gemini-3.5-flash-lite`, at version 8, whose outgoing request omits the reasoning-budget argument because that provider returns an invalid-argument error when the argument is present at all. That is a difference in the transport contract rather than evidence about the model, it is recorded in that route’s own manifest, and the amendment introducing per-route contracts is dated in the collection protocol. For every other route the outgoing request is byte-identical to the earlier runs. One failed run is retained rather than deleted: the same route at task version 7 ended in exhausted transport retries and produced no scoreable answer, so it appears nowhere in the table and is kept as a transport failure beside the campaign.

*Two admitted routes pass by nothing at all.* State reconstruction and terminal scoring are each scored over fifteen answers, so a 75% threshold asks for 11.25 correct and the lowest score that can clear it is 12 of 15, or 80.0%. GPT-5.5 clears state reconstruction at exactly that score and Claude Sonnet 5 clears terminal scoring at exactly that score, so both are admitted with a margin of zero. That is one reason a verdict is used here as a declared entry condition and never as a trait estimate. No analysis in this paper interpolates between probe scores, ranks admitted routes by probe accuracy, or reads the admitted against refused split as a measured contrast.

*The split follows the tier word in the route name.* All four refused routes carry `mini`, `nano` or `lite` as a whole word of their name, and none of the five admitted routes does. On this roster, then, the screen draws no line that the route name does not already draw. Whether a screen of this kind measures comprehension at all is examined in a concurrent submission [3], which also reports how its verdicts hold up and how it compares with simpler signals. This supplement uses the screen only as the entry condition it is.

*Nothing was discarded.* All nine routes played the same confirmatory baseline with the same race counts, and every refused route’s behaviour is reported in this supplement beside the five admitted ones. Four further advertised routes never produced a scoreable

answer at all, because transport failed before any probe was sampled; they were retained as failure records and never replaced by a substitute endpoint.

## A.2 A second, larger battery on the one model with published weights

One route in this study is not a hosted endpoint and is not one of the nine. Its weights are published, so decoding is under our control rather than a provider’s, and it is the route used for the paired decision-card diagnostic described at the end of this subsection. It was screened by its own instrument, a battery of 41 probes over the same six domains, and that battery is reported here so that the two are never mistaken for one another. Nothing in this subsection enters a strategic claim, and no score here is pooled with a score from the nine-route screen: different probe bank, different route, different number of repetitions.

The model is Qwen2.5 7B Instruct at half precision. Probes were put three ways rather than one: unaided, reworded into a paraphrase that keeps the question and changes the sentence, and accompanied by a disclosed calculation of the quantities the question needs. Each probe was answered five times in each of the three, at temperature zero under one fixed base seed, and the seven probes that list their answer options were additionally asked with those options in the reverse order. That gives 685 retained probe answers, every one of which the semantic scorer could read.

What the battery shows is a gap between stating a rule and using it. Table 3 reports it by domain. Recalling the rules and reading the stage-payoff matrix are near perfect unaided, and everything that requires carrying the state forward or doing the arithmetic collapses: unaided, this route rebuilds the current state on about a quarter of its answers, and takes a single state transition or computes a closed-form expected payoff on none of them. Disclosing the arithmetic lifts the whole battery from 52.1% to 75.6%, which measures the use of a disclosed result and not unaided comprehension. Two further readings follow. Unaided, every one of these scores sits below the matching condition the nine-route screen declares, and while the two instruments differ enough that this is a description rather than a verdict, it is the reason this route supplies a presentation diagnostic

**Table 2: The nine screened routes on the three conditions that decide admission, and which condition refused each route that did not enter. A route is admitted only when all three clear: overall accuracy at least 80%, state reconstruction at least 75% and terminal scoring at least 75%. Every cell is the percentage of that domain’s probes answered correctly. Overall accuracy rests on all 60 scored answers per route, state reconstruction and terminal scoring on 15 each, so those two move in steps of one fifteenth, about 6.7 points. Expected payoff is recorded as a diagnostic and gates nothing; no route reaches 75% on it. Rule recall, payoff reading and state transition are not admission conditions and are not shown.**

| Exact route                          | Overall accuracy | State recon. | Terminal scoring | Verdict                 |
|--------------------------------------|------------------|--------------|------------------|-------------------------|
| google/gemini-3-flash-preview        | 93.3             | 93.3         | 100.0            | admitted                |
| anthropic/claude-opus-5@default      | 91.7             | 93.3         | 100.0            | admitted                |
| openai/gpt-5.4-2026-03-05            | 90.0             | 100.0        | 100.0            | admitted                |
| openai/gpt-5.5-2026-04-23            | 90.0             | 80.0         | 100.0            | admitted                |
| anthropic/claude-sonnet-5@default    | 85.0             | 100.0        | 80.0             | admitted                |
| google/gemini-3.1-flash-lite-preview | 80.0             | 73.3         | 80.0             | refused, state recon.   |
| openai/gpt-5.4-mini-2026-03-17       | 75.0             | 66.7         | 86.7             | refused, two conditions |
| google/gemini-3.5-flash-lite         | 65.0             | 60.0         | 60.0             | refused, all three      |
| openai/gpt-5.4-nano-2026-03-17       | 51.7             | 20.0         | 66.7             | refused, all three      |

**Table 3: The 41-probe battery on the one model with published weights, Qwen2.5 7B Instruct, by domain. “Unaided” pools the plain and the reworded askings and both option orders; “disclosed” is the same probes accompanied by the arithmetic they need. Every cell is the percentage of answers the semantic scorer marked correct, over the number of answers shown. This battery is a separate instrument from the nine-route screen and its scores are never compared with those in Table 2.**

| Domain                  | Unaided | <i>n</i> | Disclosed | <i>n</i> |
|-------------------------|---------|----------|-----------|----------|
| Rule recall             | 96.4    | 140      | 100.0     | 50       |
| One-stage payoff lookup | 100.0   | 40       | 100.0     | 20       |
| State reconstruction    | 27.8    | 90       | 55.6      | 45       |
| State transition        | 0.0     | 60       | 66.7      | 30       |
| Terminal scoring        | 45.5    | 110      | 75.0      | 40       |
| Expected payoff         | 0.0     | 40       | 50.0      | 20       |
| All probes              | 52.1    | 480      | 75.6      | 205      |

and nothing else. With the arithmetic disclosed the route reaches the terminal-scoring condition exactly and still clears neither of the other two. And strict format compliance, 32.1% over the whole battery, is far below the semantic parse rate, which was 100%, so on this route the format a reader sees and the answer the model gives come apart almost two times in three.

The claim this battery can carry is narrow and worth stating as the analysis records it: probe accuracy supports rule and arithmetic performance under the tested prompt pool, it does not establish an internal world model, and accuracy with the calculation disclosed measures the use of disclosed outputs rather than unaided comprehension. The gameplay beside it is the paired diagnostic itself, 30 races and 558 decisions in each of the two presentations with no parse failure, and it is read only as a within-route contrast between two presentations of one mechanism.

### B. Robustness of the main two-player results.

### A.3 Earlier two-route confirmatory baseline

The new neutral gameplay baseline used protocol v3, the same mechanism and prompt for the two routes used in the direct EGT comparison, ten repetitions per risk level, and 30 races per route. Each route produced 558 dependent within-race decisions with zero parse failures. The entries in Table 4 are decision-weighted UNSAFE rates; they are not pooled across routes and do not estimate an evolutionary equilibrium.

**Table 4: Two-route confirmatory block used for the direct EGT comparison. Each route has 30 races, 558 dependent within-race decisions, and zero parse failures.**

| Exact route                       | <i>r</i> = .1 | <i>r</i> = .6 | <i>r</i> = .9 | Races |
|-----------------------------------|---------------|---------------|---------------|-------|
| google/gemini-3-flash-preview     | 98.9%         | 73.1%         | 60.2%         | 30    |
| anthropic/claude-sonnet-5@default | 89.2%         | 48.4%         | 32.3%         | 30    |

*Race-weighted sensitivity.* The headline entries above weight each decision equally, matching the run summaries. Because both seats in a race share the same horizon draw, we also report the race as the independent unit. Table 5 repeats the risk-response contrast for the five admitted routes using equal weight per race and a percentile bootstrap over race clusters. The point estimates preserve the same route ordering and the intervals are a sensitivity check on weighting, not an additional model run.

**Table 5: Risk-response sensitivity to the weighting unit. The contrast is the decision-weighted or race-weighted UNSAFE rate at maximum private risk 0.1 minus the rate at 0.9. Race-weighted brackets are 95% percentile bootstrap intervals over ten race clusters per risk cell.**

| Admitted route  | Decision-weighted | Race-weighted [95% RI] |
|-----------------|-------------------|------------------------|
| Gemini 3 Flash  | 38.7              | 39.1 [35.8, 42.2]      |
| Claude Opus 5   | 100.0             | 100.0 [100.0, 100.0]   |
| GPT-5.4         | 40.3              | 40.7 [34.1, 47.4]      |
| GPT-5.5         | 38.2              | 39.7 [33.1, 46.9]      |
| Claude Sonnet 5 | 57.0              | 55.0 [50.4, 59.8]      |

The Gemini manifest records that the hosted SDK stripped the requested sampling seed. The Claude manifest records that the seed was forwarded but provider-side application was not confirmed. Temperature was requested as 0.7 for both routes but was not forwarded by this SDK path. These facts bound the result to the exact hosted route and software contract.

To make the comparison boundary explicit, Figure 1 separates the risk response, the route-specific departure from the strong-selection reconstruction, the nearest-rule diagnostic, and its mismatch rate. This is an analysis artifact rather than an equilibrium test: the frontier routes generate prompted self-play trajectories, whereas EGT samples stationary population composition.

Which strategy takes that population is a one-line result, now shown in Figure 2. Read off the archived EGTtools fixation probabilities, the strategy that dominates the independently simulated finite-mutation chain ( $\beta = 2$ ,  $\mu = 0.02$ ) switches with risk: Always UNSAFE at  $p_r^{\max} = 0.1$  with 91.9% of the chain, the conditional rule that opens UNSAFE at 0.6 with 96.3%, and the conditional rule that opens SAFE at 0.9 with 93.9%. Exactly nine of the twelve off-diagonal invasions per risk level clear neutral drift  $1/Z = 0.01$ , and the small-est fixation probability among those nine is 0.997, so the ordering is not marginal. Two boundaries travel with that sentence. The fixation probabilities are evaluated in the EGTtools small-mutation limit while the shares come from the finite-mutation chains, which are two different simulation layers; and no language-model route plays this population process, so none of it is a latent-strategy statement about any endpoint.

#### A.4 The shape of a policy behind a rate

Every gameplay number in §A.3 is a rate: the share of decisions that came out UNSAFE. A rate is silent about the thing an iterated game is for. Two routes can sit at the same rate while one is reading the board and the other is reciting a constant, and no table in this supplement separates those two cases. Figure 3 is the object that does, and that is what it adds to the neutral baseline: not another number for how much UNSAFE play there was, but the policy that produced it.

The read-out is the memory-one fingerprint of the evolutionary game theory literature: the probability of playing UNSAFE given the pair of moves that preceded it, the route’s own move first and its rival’s second. The choice is not an analytical convenience. The prompt shows the players only the immediately preceding round, so memory-one is the whole state the game makes available, and four conditional probabilities are therefore a complete description of a stationary policy rather than a summary of one. The fingerprint is computed over the same nine-route neutral baseline the rest of this paper reports, 270 races and 5,022 decisions, of which the 4,482 that have a predecessor round enter the calculation; round one is excluded rather than folded in, because an opening move is a different object from a reply.

The two lagged effects beside it are Mantel-Haenszel differences, stratified on the risk level and on the other lagged move. Both stratifications are load-bearing. Risk has to be held fixed because the three risk cells have very different base rates, so pooling them would let a route that simply plays more UNSAFE at low risk look responsive. The other lagged move has to be held fixed because

these are self-play races: both companies are the same route, so an unstratified reading of what a route does after its rival went UNSAFE would largely measure the route’s own phase. Intervals resample races within their risk cell, the same clustering every other interval in this paper uses.

Reading the two panels together separates three kinds of route that a rate column would print side by side. Six of the nine answer their rival’s last move by moving toward it, and every route whose terms are defined at all has a negative own term, so the pattern is not momentum in either direction but matching the rival while correcting itself. On GPT-5.4 mini that own term is the one whose interval spans zero,  $-4.8$  pp  $[-12.7, +3.4]$ , so seven of the eight are estimated away from zero and the eighth is not. Two routes, GPT-5.4 mini and GPT-5.4 nano, carry a rival term pointing the other way; they are also the two whose UNSAFE rate barely moves with the stated risk. Claude Opus 5 is the third kind: a step whose two seats never diverged in 279 rounds, for which the question the fingerprint asks does not arise.

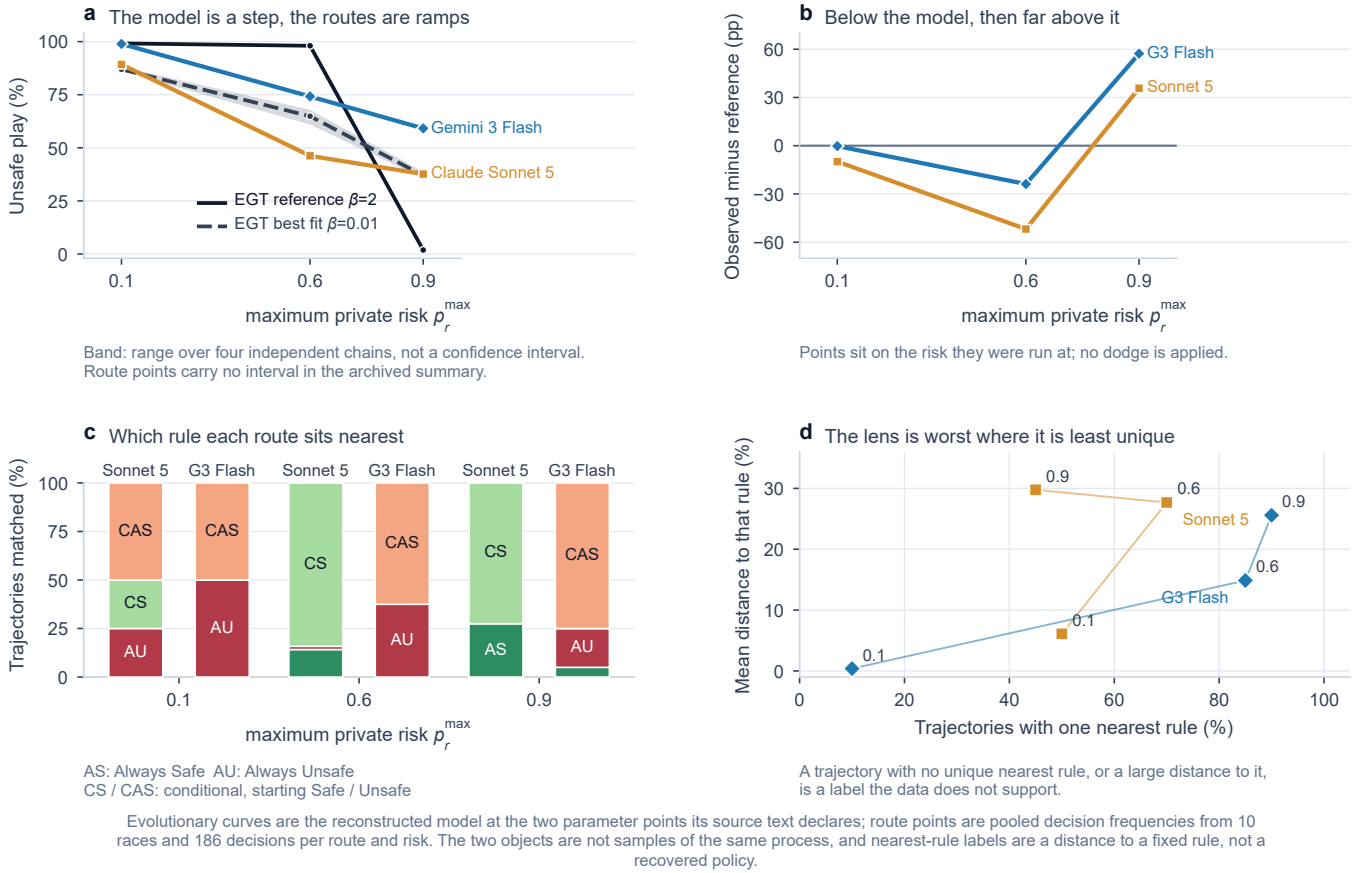
*What it does not show.* Responding to a rival is not understanding a rival, and nothing here licenses a claim about strategy, intent or reasoning: these are conditional frequencies in one game at one prompt version, on nine commercial endpoints that are not a sample from a population of models. The own term in particular is not a preference. After its own UNSAFE move a route is also carrying more accumulated private risk, which is a real feature of the game rather than a confound, but it means the term measures a response to the state and not to the move alone.

#### A.5 How much of a route’s number is the run?

Every route-level rate in this paper comes from one thirty-race run, so a reader is entitled to ask how much of a rate is the route and how much is the particular run. One route answers that question directly. Sampling on this route is not reproducible. On the three Google routes the SDK strips the seed request before it reaches the provider, and every run manifest for those routes records that. On the OpenAI and Anthropic routes the SDK forwarded the seed, and no provider confirmed applying it. Two runs of the same Google cell are therefore genuinely independent draws from the route, not a rerun of one draw.

google/gemini-3-flash-preview was collected twice under the frozen baseline protocol, version 3: at task version 3 on 2026-09-08, which is the run reported everywhere else in this paper, and again at task version 4 on 2026-09-09 on a different billing identity. The repeat was not planned as evidence. It rode along with the push that carried the reasoning-budget amendment of §A.1 and was recovered from the run archive afterwards. Protocol, prompt hash, mechanism, seed structure and the full set of thirty (game seed, risk) blocks are identical across the two runs, and the released analysis code refuses the comparison unless they are.

Agreement is close. UNSAFE rates move from 98.9% to 100.0% at  $p_r^{\max} = 0.1$ , from 73.1% to 74.2% at 0.6, and from 60.2% to 59.7% at 0.9: the largest per-risk difference over 186 decisions is 1.1 percentage points, and the risk response, the UNSAFE rate at risk 0.1 minus the rate at 0.9, moves from 38.7 to 40.3 percentage points. That is small beside the spread in risk response across the nine routes of the neutral baseline, which runs from 7.0 points on GPT-5.4



**Figure 1: The reduced evolutionary model against the two routes that were run through the comparison protocol, anthropic/claude-sonnet-5@default and google/gemini-3-flash-preview. These are two of the five routes the quality-control screen admits, not the admitted set. (a) At its reference parameter point the model is a step in risk, holding UNSAFE play near the ceiling through  $p_r^{\max} = 0.6$  and dropping it to 1.9% at 0.9; both routes are ramps and both stay far above the model at 0.9. The band on each model curve is the range across four independent seeded chains, which is a convergence diagnostic and not a confidence interval; the route points carry no interval because the archived summary holds none. (b) The same contrast signed against the reference, with each point on the risk it was run at. (c) Which of the four rules each route’s trajectories sit nearest, one stack per route. (d) Why (c) is a lens and never a finding: the labels are least unique exactly where the distance to the nearest rule is largest.**

mini to 100.0 points on Claude Opus 5, a spread of 93.0 points. That between-route spread is the comparison this check exists to protect.

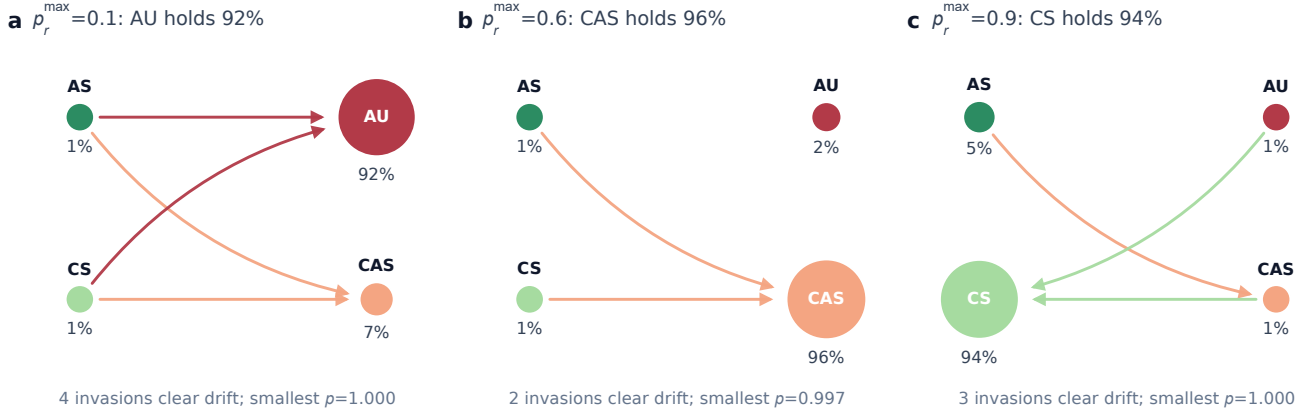
It is one route and one repeat, so it bounds run-to-run variation for this route at this sample size and licenses nothing for the routes that were collected once. The repeat is stored apart from the main campaign records on purpose. Both campaign analysers key their results by model route, so a second run of an already-represented route stored with the campaign would not raise an error. It would quietly displace the reported one.

## A.6 Selection-strength sweep of the evolutionary benchmark

The comparison in §A.3 uses one setting of the reconstructed evolutionary model. A reviewer may reasonably ask whether the gap between that model and the audited routes is a property of the

model or an artifact of that one setting. This subsection answers that question by sweeping the setting.

Two numbers govern the model. *Selection strength*  $\beta$  says how strongly a simulated agent prefers to copy a better-performing neighbour: at  $\beta$  near zero copying is almost random, and at large  $\beta$  it is almost deterministic. The *mutation rate*  $\mu$  is the chance, per update, that the sampled individual abandons its strategy and switches uniformly to one of the other three. The sweep covers ten values of selection strength,  $\beta \in \{0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 1, 2, 3, 10\}$ , crossed with three rules for setting the mutation rate:  $\mu = \beta/Z$ , the main-text convention, which gives  $\mu = 0.02$  at  $\beta = 2$ ;  $\mu = 1/Z = 0.01$ ; and a fixed  $\mu = 0.05$ , the convention under which the source study reports its best fit. With  $Z = 100$  individuals, four strategies, and the mechanism’s three maximum-private-risk levels (0.1, 0.6, 0.9), the sweep contains 90 cells built from 29 distinct parameter



Arrow: the invader fixates in the resident population with probability above neutral drift  $1/Z=0.01$ , drawn from the displaced population toward the strategy that displaces it, in the invader's colour. Disc area: share of the independently simulated finite-mutation chain ( $\beta=2$ ,  $\mu=0.02$ ), because the small-mutation transition has more than one absorbing class here and its stationary vector is not unique. No language-model route plays this process.

**Figure 2: Risk-conditioned invasion topology in the reconstructed evolutionary game. Each panel is a standard invasion graph at one maximum private-risk level. The four nodes are Always Safe (AS), Always Unsafe (AU), Conditionally Safe (CS) and Conditionally Antisocial Safe (CAS); node area shows the share of the independently simulated finite-mutation population. An arrow points from the resident strategy to an invader whose EGTtools fixation probability exceeds neutral drift  $1/Z = 0.01$ . The graph shows which strategy can replace which resident, while node areas show the finite-mutation population composition.**

pairs, since some pairs coincide across rules and were simulated once. Each cell is estimated from four independent chains of 400,000 steps with a burn-in of 100,000, thinned by 100, giving 4,000 retained samples per chain.

The reference setting reproduces. At  $\beta = 2$  with  $\mu = \beta/Z = 0.02$ , the sweep returns a decision-weighted UNSAFE share of 99.2%, 98.0%, and 1.9% at maximum risks 0.1, 0.6, and 0.9: the same three-point profile the main paper attributes to the published reference, matched to within 0.0002, 0.0001, and 0.0001 in share, far inside the 0.02 tolerance the check declares in advance. The sweep uses a deliberately disjoint seed scheme from the reproduction script, so this agreement is an independent-seed check rather than a rerun of the same random numbers.

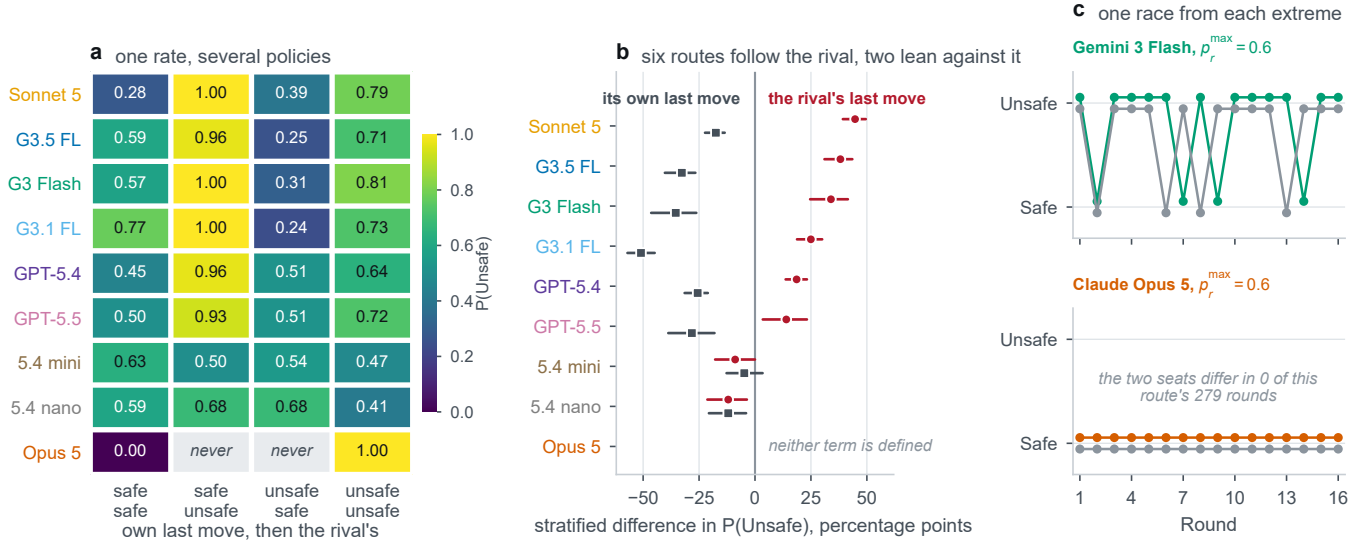
The qualitative gap does not close anywhere on the grid at the reference selection strength, but it does close at very weak selection. Of the 90 cells, four reproduce the qualitative shape the admitted endpoints show, namely near-total UNSAFE play at risk 0.1, a middling rate at 0.6, and a lower but clearly non-zero rate at 0.9, declining at every step. Three of those four also pass the mixing diagnostic. All of them sit at weak selection:  $\beta = 0.01$  and  $\beta = 0.03$  under  $\mu = 0.05$ , and  $\beta = 0.003$  and  $\beta = 0.01$  under  $\mu = 1/Z$ . Ten of the 90 cells are flagged as mixing-suspect, meaning their four independent chains disagreed by more than 0.05 on the aggregate UNSAFE share, so their chain mean must not be read as a stationary distribution; almost all of them are the very small mutation rates that  $\mu = \beta/Z$  produces at small  $\beta$ , which make the four uniform-strategy states nearly absorbing.

The best-fitting cell is the same for both audited routes, and it is not the reference setting. Fitting is by root-mean-squared deviation, in percentage points, between the model's three-point UNSAFE profile and the route's observed decision-weighted UNSAFE rates

at the same three risk levels; the largest single-condition deviation is reported beside it so that one badly missed condition stays visible. For both routes the minimum-error cell is  $\beta = 0.01$  with fixed  $\mu = 0.05$ , whose profile is 87.3%, 63.9%, and 38.0%. Against anthropic/claude-sonnet-5@default, observed at 89.2%, 48.4%, and 32.3%, that cell has an error of 10.28 percentage points and a largest single-condition miss of 17.70 points, at risk 0.6. Against google/gemini-3-flash-preview, observed at 98.9%, 73.1%, and 60.2%, the error is 15.13 points and the largest miss 21.14 points, at risk 0.9. Even the best cell on the grid is therefore more than ten percentage points away from either route.

*All-five admitted-route extension.* The preceding block is the earlier two-route confirmatory comparison. To check whether its qualitative reading depends on that restricted roster, we fit the same existing 90-cell sensitivity table to the canonical neutral-baseline profiles of all five admitted routes. This is analysis only: no LLM requests or evolutionary chains were added. Distances are RMSEs in percentage points over the three matched risk levels, and the primary cells below are required to be well mixed. The best weak-selection cell is the same in every route,  $\beta = 0.01$  with fixed  $\mu = 0.05$ ; the four graded routes are substantially closer to it than to the reference  $\beta = 2$  (Table 6), while Claude Opus 5 remains a qualitative near-step exception.

One boundary is worth restating, because the reproduction lane states it itself. This is a repo-native reconstruction of the disclosed reduced evolutionary model, not a bitwise reproduction of the original authors' code: that code, the generated payoff matrices, the payoff Monte Carlo seeds, and the exact library revision are not public in the preprint version we worked from, and the compiled comparison class could not be executed on this interpreter because



**Figure 3: The memory-one policy behind each route’s UNSAFE rate, over the nine-route neutral baseline. (a) One row per route and one column per preceding pair of moves, the route’s own first and its rival’s second, with the conditional probability printed in every tile; row labels carry the route colours used throughout the paper, so the panel needs no legend. Two of Claude Opus 5’s tiles are marked absent rather than left blank: its two seats never once played differently, so the mixed contexts did not arise and there is nothing there to measure. (b) The two lagged associations on one scale, Mantel-Haenszel differences in  $P(\text{UNSAFE})$  stratified on the risk level and on the other lagged move: how far UNSAFE play differs after the rival’s UNSAFE move, and how far it differs after the route’s own, with race-clustered 95% intervals and the routes in the same order as panel a. (c) Two races at the same risk level under the same protocol, one from a route that answers its rival and one from the route that does not move at all, with the two seats drawn separately. Every quantity here is an association within a stratum and not a causal effect. Nothing is randomised: these are conditional frequencies from nine commercial endpoints in one game at one prompt version, self-play throughout, and the own term in particular also carries the accumulated private risk that an earlier UNSAFE move leaves behind, so it is a response to the state and not to the move alone.**

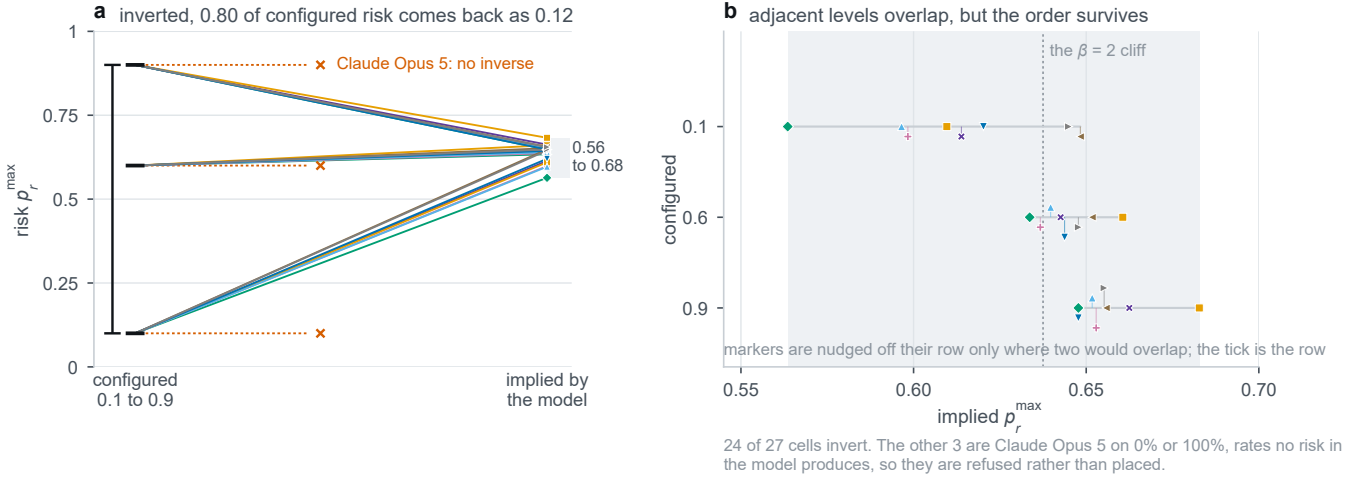
**Table 6: Analysis-only EGT fit across all five admitted routes. “Reference” is the  $\beta = 2$ ,  $\mu = \beta/Z$  cell; “weak” is the best well-mixed cell with  $\beta \leq 0.03$ . Values are RMSE in percentage points.**

| Route           | Observed shape   | Reference | Best weak |
|-----------------|------------------|-----------|-----------|
| Gemini 3 Flash  | graded decline   | 36.6      | 15.4      |
| GPT-5.4         | graded decline   | 32.9      | 5.1       |
| GPT-5.5         | graded decline   | 34.9      | 11.3      |
| Claude Sonnet 5 | graded decline   | 34.1      | 9.6       |
| Claude Opus 5   | near-step switch | 56.6      | 43.6      |

no matching wheel exists for it, so the sweep ran the pinned pure-Python implementation instead. The quantities above are therefore seeded-chain estimates reported with their between-chain spread, not exact stationary distributions, and the route rates beside them are descriptive prompted self-play endpoints rather than draws from a population process.

*Reading the model backwards.* The sweep above asks what the model predicts. The complementary question is what an observed rate would tell us about the model’s own risk parameter, and Figure 4 asks it. The answer bounds how finely this benchmark can be read at all. Because the only place the model bends is its cliff, asking which risk would make it emit an observed rate sends nearly every

answer to the same place: the three configured levels span 0.80 of the risk axis, and the 24 invertible route-by-risk cells come back inside a band 0.12 wide, a compression of about sevenfold. What is lost is resolution and not ranking, and the distinction matters. All eight invertible routes return their three implied risks in the configured order, and seven of the eight cells run at 0.1 and strictly left of every cell run at 0.9, the two extreme rows sharing only 0.0006 of the axis. But adjacent rows overlap, the 0.1 and 0.6 rows on 0.0147 and the 0.6 and 0.9 rows on 0.0129, so a rate read back through this model fixes an implied risk only to about a tenth of the axis it was configured on. The inversion runs at  $\beta = 0.01$ , the weak-selection point, which is the setting that gives the model its best chance of spreading the cells apart; at  $\beta = 2$  the inverse is defined only inside a 0.005-wide window, and inverting there would manufacture the compression. Three limits travel with the figure. It is a statement about the model, not an estimate of any route’s perceived risk: it says only what the model would have to be handed. A compressed inverse is not an unresponsive route, since all nine routes move the right way with risk and Claude Opus 5 most steeply of all; eight of them merely also land inside the model’s own range, which is what lets the inversion answer at all. And every curve behind it is the small-mutation limit, which cannot represent the finite mutation rate of the parameter points it is drawn at.



**Figure 4: Inverting the reconstructed evolutionary model through the observed UNSAFE rates of the nine audited routes, at the weak-selection point  $\beta = 0.01$ .** (a) The inversion as a slope chart: configured risk on the left at full scale, the risk the model would need in order to emit the route’s observed rate on the right at the same full scale. The fan closes: 0.80 of configured risk comes back as 0.12. Claude Opus 5 has no inverse at any level, because its observed rates of 0% and 100% are rates no risk in the model produces; those three cells are refused rather than placed, and marked as refusals. (b) That band magnified, one row per configured level, with the  $\beta = 2$  cliff marked. Adjacent rows overlap, on 0.0147 and 0.0129 of the axis, while the whole configured span of 0.80 collapses to 0.12; what the inversion loses is resolution rather than ranking, since all eight invertible routes keep the configured order and seven of the eight 0.1 cells lie strictly left of every 0.9 cell. Where two markers would fall on each other the second is nudged off its row with a leader back to it; the tick is the row. Neither panel is an estimate of a route’s perceived risk, and a compressed inverse is not an unresponsive route.

## A.7 Crossed representation-robustness design

An earlier local pilot varied the story the game is told through and the symbols used for the two actions, but varied them together with repetition parity, so the two could not be separated, and it failed its screen. It is therefore excluded, and a fully crossed rerun was required before any claim about representation. That rerun has now completed on the two routes selected for this study, and this subsection reports it.

The rerun crosses two *narrative skins*, that is two stories that leave the rules untouched, against two *symbol mappings*, that is which of the two opaque response codes  $P$  and  $Q$  stands for SAFE and which for UNSAFE, inside repetition blocks, at each of the mechanism’s three maximum-private-risk levels. That is twelve cells per route. It completed on both selected routes under one frozen protocol, the context-and-mapping protocol, version 3, each with ten repetitions, 120 races, 2,232 recorded decisions and zero parse failures. The two routes are Gemini 3 Flash and Claude Sonnet 5, whose full route strings are given in Table 2. The horizon and set-back random-number streams are held common across conditions, so two cells differ in their prompt and not in the environment they met. Table 7 gives all twelve cells for the first route; the second route’s cells are released with the analysis code, and both routes’ paired contrasts are in Table 8.

Pooling cells is not the sharpest way to read this design. The repetition index is a common-random-number block, so the same index reuses the same sampled horizon across every skin and mapping cell, and a recorded hash of the horizon draws lets that be checked rather than assumed. It holds for all 60 paired blocks on

**Table 7: Fully crossed narrative-skin by symbol-mapping rerun, first of the two routes that completed it, google/gemini-3-flash-preview; the second route, anthropic/claude-sonnet-5@default, completed the same twelve cells, and its cell rates are released with the analysis code. “Mapping” names which response code stands for SAFE: under P\_SAFE\_Q\_UNSAFE the code  $P$  means SAFE, and under Q\_SAFE\_P\_UNSAFE it means UNSAFE.  $n_{\text{race}}$  is the number of independent races in the cell and  $n_{\text{dec}}$  the number of decisions recorded in it; the UNSAFE rate is decision-weighted. All twelve cells are complete and no response failed to parse.**

| Narrative skin  | Mapping         | Risk | $n_{\text{race}}$ | $n_{\text{dec}}$ | UNSAFE actions | UNSAFE rate |
|-----------------|-----------------|------|-------------------|------------------|----------------|-------------|
| abstract_game   | P_SAFE_Q_UNSAFE | 0.1  | 10                | 186              | 186            | 100.0%      |
| abstract_game   | P_SAFE_Q_UNSAFE | 0.6  | 10                | 186              | 156            | 83.9%       |
| abstract_game   | P_SAFE_Q_UNSAFE | 0.9  | 10                | 186              | 111            | 59.7%       |
| abstract_game   | Q_SAFE_P_UNSAFE | 0.1  | 10                | 186              | 186            | 100.0%      |
| abstract_game   | Q_SAFE_P_UNSAFE | 0.6  | 10                | 186              | 186            | 100.0%      |
| abstract_game   | Q_SAFE_P_UNSAFE | 0.9  | 10                | 186              | 166            | 89.2%       |
| technology_race | P_SAFE_Q_UNSAFE | 0.1  | 10                | 186              | 186            | 100.0%      |
| technology_race | P_SAFE_Q_UNSAFE | 0.6  | 10                | 186              | 144            | 77.4%       |
| technology_race | P_SAFE_Q_UNSAFE | 0.9  | 10                | 186              | 81             | 43.5%       |
| technology_race | Q_SAFE_P_UNSAFE | 0.1  | 10                | 186              | 186            | 100.0%      |
| technology_race | Q_SAFE_P_UNSAFE | 0.6  | 10                | 186              | 185            | 99.5%       |
| technology_race | Q_SAFE_P_UNSAFE | 0.9  | 10                | 186              | 83             | 44.6%       |

both routes. Differencing inside a repetition therefore removes the horizon draw from the comparison and leaves the presentation as the only thing that varies. Table 8 reports those paired contrasts, with intervals from a percentile bootstrap over repetition blocks.

Two things stand out. The symbol mapping moves play by a similar amount on both routes, about eight to ten percentage points, even though the two routes differ by provider. The narrative skin also moves play on both, but by very different amounts, so skin sensitivity is route-specific in a way that code sensitivity is not. All four intervals exclude zero. Nothing in the mechanism differs across any of these cells: the payoff matrix, the progress increments, the hidden horizon law and the risk formula are identical, and both routes had already passed the quality-control screen. Two routes is enough to show the effect replicates and not enough to characterise endpoints in general.

Panel b of Figure 13 draws both routes’ cells as an interaction plot on one common vertical span. That figure is placed with the matched group-size comparison of §A.24, because that is where all three of this supplement’s manipulations of the same game can be read side by side. What the panel adds here is the question the table cannot answer: the two factors do not interact on either route, so the narrative skin and the symbol mapping simply add, and the crossing that this rerun was built to test returns an interval spanning zero on both routes.

**Table 8: Repetition-paired presentation contrasts on the two crossed routes. Each contrast is the mean difference in UNSAFE rate within a repetition block, which holds the sampled horizon fixed; positive means the second level raises UNSAFE play. Intervals are 95% percentile bootstrap intervals over the 60 paired blocks. Horizon pairing was verified for 60 of 60 blocks in every row. Rates pool decisions within a cell, while a contrast is the mean over matched blocks of the within-block difference in race-level rates, so a contrast need not equal the difference of the rates in Table 7.**

| Route           | Contrast            | Effect   | 95% CI        |
|-----------------|---------------------|----------|---------------|
| Gemini 3 Flash  | code $Q$ means SAFE | +9.5 pp  | +5.6 to +13.8 |
| Claude Sonnet 5 | code $Q$ means SAFE | +8.3 pp  | +5.4 to +12.2 |
| Gemini 3 Flash  | abstract contest    | +10.8 pp | +6.0 to +15.6 |
| Claude Sonnet 5 | abstract contest    | +3.4 pp  | +0.9 to +7.4  |

Both routes’ marginal cells and both paired contrasts are regenerated by the released analysis code, which refuses a route whose race count, ten-per-cell balance or parse-failure count does not check out. It writes both routes’ cell rates with a provenance record beside them. An earlier attempt on the Claude route stopped at 106 of 120 races on a provider quota refusal. It is retained as a failure record that points to the complete rerun reported here, and none of its cells is reported, because a counterbalanced design read at 106 of 120 races is unbalanced rather than merely smaller.

Two things are visible in Table 7 and neither is a general finding. First, the risk response survives both manipulations: in every one of the four skin-by-mapping combinations the UNSAFE rate is 100% at risk 0.1 and lower at risk 0.9. Second, the mapping is not inert. Holding the skin fixed, swapping which code means SAFE moves the rate at risk 0.9 from 59.7% to 89.2% under the abstract story and from 43.5% to 44.6% under the technology story, and at risk 0.6 from 83.9% to 100% and from 77.4% to 99.5% respectively. A relabelling

that a correct model of the game should ignore therefore changes the action sequence.

What follows from this table is bounded by how many routes stand behind it. The design is crossed on both routes selected for the crossed representation study, Gemini 3 Flash and Claude Sonnet 5, whose full route strings are in Table 2, each at the full twelve cells, so the mapping effect is not confined to one provider’s decoder at one revision: it appears on both, at +9.5 and +8.3 points. Two routes are still two routes. They do not separate what belongs to this task from what belongs to commercial endpoints in general, they are not a sample from which a population of models could be described, and the narrative skin already behaves differently on the two, so a third route could as easily break the pattern as extend it. We therefore report the symbol-mapping effect as replicated across the two audited routes, and make no claim about symbol mapping or narrative skin beyond them.

## A.8 Human-benchmark replication check

Before comparing model-generated behaviour with the human benchmark of Domingos and Han [5] in the main paper’s diversity comparison, we first validated our reconstruction of that benchmark by refitting its published dynamic specification on the de-identified dataset of Domingos and Han [5]. The reconstructed estimation sample contains 2,888 round-level observations, 172 race-pair clusters, and 338 participants. Table 9 compares the refit coefficients against the values reported in the original study.

The human reference flow is 341 released participants, 340 with a complete first-five-round action history, and 338 with complete values on every covariate in the published dynamic specification. The raw sequence comparison therefore uses 340 participants; the descriptor and elicitation analyses use 341 because they can be computed for one further participant whose complete sequence is unavailable; and the dynamic refit uses 338. No participant is added anywhere, and each figure is a subset of the same released dataset.

**Table 9: Replication of Domingos and Han [5]’s dynamic specification on the de-identified dataset of Domingos and Han [5]. “Reconstructed” is our refit; “Original” is the coefficient reported in the source study.**

| Term                         | Reconstructed | Original |
|------------------------------|---------------|----------|
| Opponent’s previous action   | 0.606         | 0.607    |
| Player’s progress gap        | −0.295        | −0.296   |
| First-round action           | 0.217         | 0.217    |
| Player’s own previous action | −0.195        | −0.193   |

## A.9 Claim-scoped citation audit

Table 10 records both the supported use of each cited study and the nearest excluded overclaim. These evidence boundaries do not turn the diagnostic pilots in this paper into confirmatory results.

The reconstruction reproduces the original coefficients almost exactly (largest discrepancy 0.002), which is the basis for treating the human-side comparisons in the main paper’s diversity comparison and the elicited-risk and persona results of §A.11 as resting on a

**Table 10: Source-level audit for citations added to the prompt-sensitivity and behavioural-validity argument.**

| Source                        | Supported use                                                                                                                                   | Excluded overclaim                                                                   |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Sclar et al. [16]             | Meaning-preserving few-shot formatting changes can yield large, model-dependent performance spreads.                                            | Not evidence for semantic paraphrase effects or a universal effect size.             |
| Pezeshkpour and Hruschka [13] | Reordering multiple-choice answer options can change predictions.                                                                               | The proposed uncertainty-plus-placement mechanism is not causally identified.        |
| Wei et al. [17]               | Order and response-token identity are distinct selection-bias targets.                                                                          | Not proof that every task or model exhibits the same bias.                           |
| Salinas and Morstatter [15]   | Controlled output-format and atomic whitespace variants can alter answers.                                                                      | Not a claim that every added space changes behaviour.                                |
| Zheng et al. [19]             | Across their factual-question setup, average persona benefit was absent or slightly negative and reliable persona selection remained difficult. | Not a universal null effect for persona-driven agent tasks.                          |
| Aher et al. [1]               | Three of four behavioural findings were qualitatively reproduced, while the fourth showed a hyper-accuracy distortion.                          | Not evidence that LLM samples are interchangeable with human populations.            |
| Akata et al. [2]              | Prediction before action improved coordination and scores in repeated games.                                                                    | Not a general claim that an observed action reveals a correct opponent belief.       |
| Herr et al. [8]               | Action-label order and payoff reassignment changed choices in two canonical two-player games.                                                   | Not evidence that every strategically equivalent transformation has the same effect. |
| Robinson and Burden [14]      | Procedurally varied vignettes produced substantial contextual variability under one underlying Prisoner’s Dilemma structure.                    | Not a repeated-game result and not an estimate for the present race.                 |
| Wu et al. [18]                | Performance declined consistently across 11 tasks when default mappings were replaced by specified counterfactual alternatives.                 | Not proof that the tested models lacked all abstract reasoning ability.              |
| del Rio-Chanona et al. [4]    | LLM markets recovered broad feedback-condition dynamics but displayed less strategy heterogeneity than human participants.                      | Not evidence that LLM agents and human participants are exchangeable.                |

correctly processed benchmark rather than on a reimplementa-  
tion error.

### C. Scripted-rival intervention.

## A.10 Five audited routes against a fixed and undisclosed rival policy

Every gameplay result reported elsewhere in this paper is self-play: both companies in a race are the same endpoint. That design cannot separate a route responding to its rival from a route running through a phase of its own, because the rival is the route. It also leaves this paper’s two lanes disconnected. The evolutionary lane is built on four reduced strategies, *Always Safe*, *Always Unsafe*, *Conditionally Safe* and *Conditionally Antisocial Safe*, and until this campaign the empirical lane had never played against any of them.

*What was collected.* Five gate-admitted routes, Gemini 3 Flash, Claude Opus 5, GPT-5.4, GPT-5.5 and Claude Sonnet 5, whose full route identifiers are in Table 2, each played the same game against each of those four strategies at each of the three private-risk levels. That is 60 cells, ten repetitions each, 600 races and 5,580 route decisions, 93 route decisions per cell, with zero parse failures and zero deviations by any scripted rival. The rival is executed by the task file rather than called, so it costs no requests and cannot be wrong about its own strategy for any reason a model could be.

Playing a language model against fixed strategies is established practice rather than something introduced here: Akata et al. [2] run always-cooperate, always-defect and defect-once agents against

five models, and Payne and Alloui-Cros [12] run Tit-for-Tat, Suspicious Tit-for-Tat and Grim Trigger, among other strategies, against frontier models under a varying termination probability. Our two conditional rivals are their Tit-for-Tat and Suspicious Tit-for-Tat. What is particular to this campaign is that the rival is never disclosed, that its every move is replayed and checked after the fact, that the route’s seat is counterbalanced, and that the same frozen design is run on more than one endpoint.

Three properties of the design are load-bearing and were asserted offline before any of it was pushed. The prompt is byte-identical to the neutral baseline’s and never says the rival is scripted, so a race here is comparable with a race there; the rival is a strategy, not a disclosure. The route’s seat is counterbalanced, five repetitions in each seat per cell, because this study has measured a seat effect in the neutral baseline, on prompts that differ only in which of two names is whose, and a fixed seat would fold that effect into every number. And a conditional strategy answers the *route*’s previous move rather than its own, which is the property that separates a rival from a mirror: a strategy answering its own history would be a constant, and the whole conditional arm would measure nothing.

*How the rival was audited.* A scripted rival is code, so it cannot be audited by reading a response. It was audited by replay: the released ingestion code recomputes every rival move from the route’s own recorded moves and refuses the cell on a single deviation. All 60 cells replayed clean. This is the one failure that would have left the data looking perfectly healthy while answering a different question than the one asked.

*The collection record.* The workload was partitioned into the analysis unit itself, one route by rival by risk cell, and each cell was collected whole on one declared account. The assignment was fixed before the final extension began and is recorded in collection receipts that are kept outside the paper, because they name the accounts. All 60 cells carry the same scripted-rival protocol and prompt version; the three task hashes are one per risk level and are identical across routes. This keeps the game mechanism fixed while allowing account rotation across cells. The account is a billing boundary, not a model boundary: the analysis never treats account identity as evidence about behaviour, and failed quota or authentication attempts contributed no races to the admitted set. The complete assignment audit is kept with those receipts. The full route-by-risk record is in Table 11; Figure 5 shows the same response and context contrasts as categorical tiles.

*Coverage and decoding contracts.* The final campaign is complete: all five admitted routes have all 12 route-by-strategy-by-risk cells. Every cell has ten races, 93 route decisions, zero parse failures, zero rival replay deviations and five races in each route seat. The routes are not homogeneous in decoding contract. Gemini 3 Flash, GPT-5.4 and Claude Sonnet 5 use the legacy 256-token, no-reasoning contract; GPT-5.5 uses a 1024-token, low-reasoning contract; Claude Opus 5 uses the latter contract in eleven cells and the legacy contract in one. Coverage therefore supports the five-route descriptive comparison, but not a claim that all endpoints were sampled under one identical decoding configuration. The 36 legacy collection receipts also omit a copied source hash field; their run manifests carry the correct risk-specific hashes, so this is a provenance metadata limitation rather than a gameplay failure.

*One-cell contract sensitivity.* The one Claude Opus 5 cell collected under the legacy decoding contract is *Always Safe* at risk 0.6. A predeclared harmonisation attempt under the active contract returned a quota 403 before any race or model output was produced, so it is retained as an infrastructure failure and the existing cell is not replaced. Excluding it as a read-only scope check removes one *Always Unsafe* – *Always Safe* contrast and one four-strategy ordering cell. The remaining evidence still has positive rival contrasts in 14/14 cells and the weak ordering in 13/14 cells (strict ordering in 11/14). This is a scope sensitivity, not a harmonised replication; full-grid values remain reported under their recorded contracts.

*Finding one: the shape is largely shared, with one boundary reversal.* Every route generally plays *UNSAFE* more often as the scripted rival becomes more *UNSAFE*, and the *Always-Unsafe* minus *Always-Safe* contrast is positive in all fifteen route-by-risk cells. The smallest interval lower bound is +16.5 points on Claude Opus 5 at risk 0.1. The weak ordering  $\text{Always Safe} \leq \text{Conditionally Safe} \leq \text{Conditionally Antisocial Safe} \leq \text{Always Unsafe}$  holds in fourteen of fifteen cells and is strict in twelve. The one reversal is GPT-5.5 at risk 0.1, where *Conditionally Antisocial Safe* reaches 100.0% and *Always Unsafe* reaches 97.8%. Both values are close to the ceiling, so the reversal is a saturated boundary observation rather than evidence against the broader direction.

The response to stated risk survives once the rival is held fixed. Against *Always Safe*, the rate never rises with risk on any of the five routes. The paired risk contrast ranges from +10.4 to +69.9

points, with Claude Opus 5 providing the largest change and the other four routes showing more graded changes. This contrast is not interchangeable with the rival contrast: both are paired within repetitions, but they change different parts of the interaction.

*Finding two: the magnitude is checkpoint-specific.* The rival contrast ranges from +25.3 points for GPT-5.5 at risk 0.1 to +87.0 points for Claude Opus 5 at risk 0.6. Gemini 3 Flash remains near +70 points across risks, GPT-5.4 remains near +50, GPT-5.5 grows from +25.3 to about +66, and Claude Sonnet 5 falls from about +66 to +45 at the highest risk. The route’s level also differs. Against *Always Safe*, GPT-5.4 plays *UNSAFE* on 36.6%, 32.3% and 25.8% of its decisions, the highest rate of the five routes at risks 0.6 and 0.9. At risk 0.1 GPT-5.5 (72.0%) and Claude Opus 5 (63.4%) are higher. Claude Opus 5 then falls to the floor at the two higher risks. These are separate endpoint samples, so non-overlapping intervals are a conservative sign of a difference rather than a test of one.

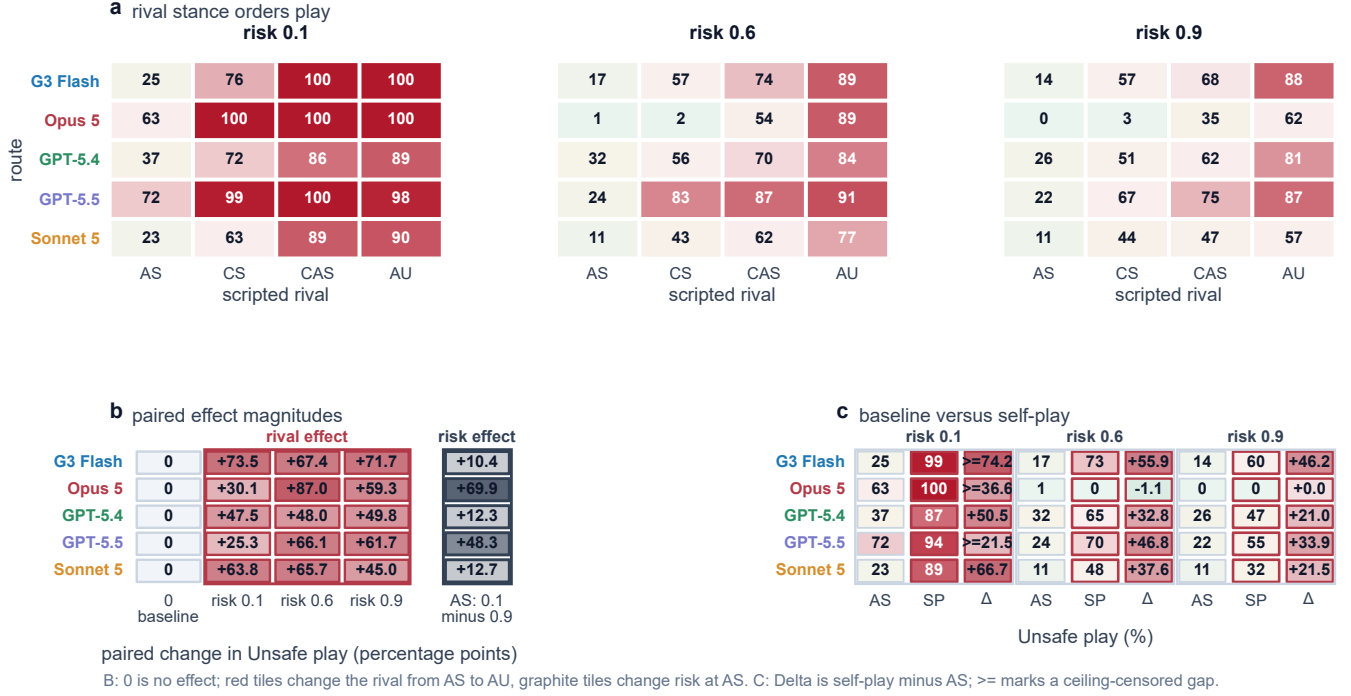
*We name the contrast for what it compares.* Calling the safe-rival arm exploitation would assert that the route takes the opportunity a non-punishing rival offers, and the measured rates say the opposite: on every route that arm carries the lowest rates anywhere in the campaign, below what the same route plays against itself. What the routes do is match the rival, keeping most of their restraint against a rival that keeps its own and abandoning much of it against one that never had any. A strategy name is not a summary of what the rival played either. *Conditionally Safe* opens *SAFE* and then mirrors; its own *Unsafe* rate varies with the route because it follows the route’s previous move. Across the five routes it played *UNSAFE* on approximately 31% to 73% of its own moves, so the strategy label should not be read as a claim about its realised average action.

*The rival’s opening move.* *Conditionally Safe* and *Conditionally Antisocial Safe* are the same strategy from round two onward, both repeating the route’s previous move, so their difference isolates the rival’s first move and nothing else. The full opening-move contrast is retained in the derived route tables. It is not used as the headline result because its interpretation is conditional on the route’s own first move, which varies substantially across the five routes.

The route’s own opening move is not a constant across the campaign: Gemini 3 Flash opened *UNSAFE* in all 120 races, Claude Opus 5 in 41, GPT-5.4 in 116, GPT-5.5 in 115, and Claude Sonnet 5 in 23. The opening-move contrast is about the rival’s first move, which the task file fixes, and not about the route’s.

*What it means for the rest of this paper.* A self-play rate is not a measurement of how a route treats risk. Self-play exceeds the fixed-safe rate in thirteen of fifteen cells. The two exceptions are Claude Opus 5 at the floor. At risk 0.6 its self-play rate is lower (0.0% against 1.1%), and at risk 0.9 the two designs are tied at 0.0%. The risk-0.1 Gemini self-play distance is a lower bound, because nine of its ten mirror-match races have every decision *UNSAFE* and that arm has no room above it. A self-play rate is an interaction outcome of one policy meeting a copy of itself, and every self-play number in this paper should be read that way.

That self-play can hide behaviour is not a new observation. Lekeas and Stamatiopoulos [9] report phenomena that are, in their words, invisible in self-play, including that which agent moves first decides which equilibrium a pair reaches. The contribution here



**Figure 5: The scripted-rival campaign on five gate-admitted routes. (a) Response surfaces across scripted rivals and stated-risk levels. (b) Paired effect magnitudes: the Always-Unsafe minus Always-Safe rival effect at each risk, with the fixed-safe risk effect shown on the same percentage-point scale. (c) Always-Safe rates, self-play rates and their signed differences; a  $\geq$  label marks a ceiling-censored gap. Confidence intervals for the rival contrast are reported in Table 11.**

**Table 11: Full five-route scripted-rival campaign. Each AS/CS/CAS/AU entry is one route-by-strategy-by-risk cell with ten races and 93 route decisions; each route-by-risk row therefore aggregates four such cells, 40 races and 372 route decisions, with both seats used and zero parse failures or rival replay deviations. Rates are percentages of the route’s own decisions. “Rival stance” is Always Unsafe minus Always Safe, differenced inside a repetition; the bracketed values are its 95% percentile interval over ten blocks. “Risk at Safe” is the paired risk-0.1 minus risk-0.9 contrast with the rival held at Always Safe, and contrast columns are percentage-point differences. Rates pool decisions within a cell. A contrast is the mean over matched repetitions of the within-repetition difference in race-level rates, so a contrast need not equal the difference of the printed rates. Claude Opus 5 has one legacy decoding cell. Coverage is complete across the five routes, but decoding contracts are heterogeneous across endpoints, so this cross-route comparison is descriptive rather than a controlled comparison under one common decoding configuration.**

| Route           | $p_r^{\max}$ | AS   | CS    | CAS   | AU    | Rival stance | 95% CI       | Risk at Safe | Self-play |
|-----------------|--------------|------|-------|-------|-------|--------------|--------------|--------------|-----------|
| Gemini 3 Flash  | 0.1          | 24.7 | 76.3  | 100.0 | 100.0 | +73.5        | [67.8, 78.7] | +10.4        | 98.9      |
|                 | 0.6          | 17.2 | 57.0  | 74.2  | 89.2  | +67.4        | [57.1, 76.6] |              | 73.1      |
|                 | 0.9          | 14.0 | 57.0  | 67.7  | 88.2  | +71.7        | [65.6, 77.5] |              | 60.2      |
| Claude Opus 5   | 0.1          | 63.4 | 100.0 | 100.0 | 100.0 | +30.1        | [16.5, 42.8] | +69.9        | 100.0     |
|                 | 0.6          | 1.1  | 2.2   | 53.8  | 89.2  | +87.0        | [84.3, 89.6] |              | 0.0       |
|                 | 0.9          | 0.0  | 3.2   | 35.5  | 62.4  | +59.3        | [51.2, 66.5] |              | 0.0       |
| GPT-5.4         | 0.1          | 36.6 | 72.0  | 86.0  | 89.2  | +47.5        | [37.0, 57.2] | +12.3        | 87.1      |
|                 | 0.6          | 32.3 | 55.9  | 69.9  | 83.9  | +48.0        | [40.0, 55.0] |              | 65.1      |
|                 | 0.9          | 25.8 | 50.5  | 62.4  | 80.6  | +49.8        | [35.8, 60.5] |              | 46.8      |
| GPT-5.5         | 0.1          | 72.0 | 98.9  | 100.0 | 97.8  | +25.3        | [20.3, 30.6] | +48.3        | 93.5      |
|                 | 0.6          | 23.7 | 82.8  | 87.1  | 91.4  | +66.1        | [62.5, 69.9] |              | 70.4      |
|                 | 0.9          | 21.5 | 66.7  | 75.3  | 87.1  | +61.7        | [53.5, 69.1] |              | 55.4      |
| Claude Sonnet 5 | 0.1          | 22.6 | 63.4  | 89.2  | 90.3  | +63.8        | [54.8, 72.1] | +12.7        | 89.2      |
|                 | 0.6          | 10.8 | 43.0  | 62.4  | 77.4  | +65.7        | [62.2, 68.9] |              | 48.4      |
|                 | 0.9          | 10.8 | 44.1  | 47.3  | 57.0  | +45.0        | [42.0, 48.2] |              | 32.3      |

is narrower and specific to this setting: what self-play hides is the

*risk response* itself, the one quantity this game exists to elicit, and it hides it on every route we have run this design on.

*What it does not show.* Five routes, one game, one prompt version, ten races per cell. Five commercial endpoints are not a sample of a population of models, and four strategies are four points in a space of policies rather than a sample of rivals, so a scripted rival establishes what a route does against *these* rivals. The within-block policy contrast is identified because rival-policy assignment is fixed and exogenous. Conditional rivals can still respond mechanically to the route’s previous action, as specified by their fixed policies. This is not causal about anything outside this game. Nothing here licenses a claim about strategy, intent, or reasoning: these are conditional frequencies. It is regenerated by the released analysis code, which takes one endpoint at a time, writes one derived table per endpoint rather than pooling them, and refuses a cell that did not complete, that carries a parse failure, that is the wrong protocol, that has the wrong number of races, whose seat counterbalance is not five and five, or whose rival deviated from the strategy its directory claims.

#### D. Human comparison and trajectory diversity.

### A.11 Risk preference and assigned personas

We finally distinguish three constructs that should not be conflated: the human participants’ pre-game elicited risk preference, the mechanism’s assigned private-risk treatment, and the narrative risk persona supplied to an LLM in its prompt.

The three persona framings themselves are drawn in Figure 6.

Human participants completed an incentivised Eckel–Grossman gamble elicitation (0 = most risk-averse, 5 = least risk-averse) before playing the two-player race: a choice among six gambles, made once, before any round is played. This elicited measure has essentially no direct association with subsequent UNSAFE rate,

$$\text{slope} = -0.004, \quad r = -0.015, \quad p = 0.79, \quad n = 341,$$

where the slope is the trend across the six group means plotted in Figure 7; the participant-level slope of the same relationship is  $-0.002$ , while the correlation, the  $p$  value and the sample size above are participant-level. The difference in mean UNSAFE play between the most and the least risk-accepting human groups, about five percentage points, lies inside the corresponding uncertainty interval. This null result independently reproduces Domingos and Han’s own preregistered finding that elicited risk preference did not significantly predict UNSAFE choice [5], and stands in contrast to the tournament literature’s general finding that human risk-taking responds to competitive standing rather than remaining a fixed trait [6, 10, 11]: in this task, the round-by-round interaction, whose largest term in the dynamic refit of §A.8 is the opponent’s previous action, is a stronger organising signal than static pre-game disposition. The multiplayer position analysis is exploratory and is not used as a second independent confirmation of this two-player result.

One further place remained where an association could have survived, and it does not. The elicitation might have failed to predict *how much* UNSAFE a participant plays while still predicting *which* dynamic pattern that participant follows. It does not do that either. Comparing the elicited score across the four behavioural archetypes of §A.21, which are the  $k$ -means groups fitted to five standardised trajectory descriptors of the same human participants, returns a

null:

$$H = 4.540, \quad df = 3, \quad p = 0.209, \quad n = 341,$$

by a tie-corrected Kruskal–Wallis test, which asks whether the distribution of one variable differs across several groups without assuming a normal distribution. The effect size is  $\epsilon^2 = 0.0046$ , so the archetype label accounts for under half of one percent of the variance in the elicited score, which is negligible on any convention. No participant is excluded: all 341 human participants carry both an archetype label and an elicited score, and the elicitation is complete.

The archetype medians are identical. All four groups have a median elicited score of 2.0 on the 0 to 5 scale, with interquartile ranges of 1.0 to 2.5 ( $n = 19$ ), 1.0 to 3.0 ( $n = 170$ ), 1.0 to 3.0 ( $n = 133$ ), and 1.5 to 3.5 ( $n = 19$ ). None of the six pairwise comparisons between archetypes is significant after Holm–Bonferroni correction across all six pairs. The smallest uncorrected  $p$  is 0.079, between the two largest archetypes, which becomes 0.476 after correction; every other corrected  $p$  is above 0.88. The same conclusion follows from Dunn’s test on mean ranks and from pairwise Mann–Whitney tests, which the artifact reports side by side.

The null is not a weakness in the comparison; it sharpens it. A measured human risk preference predicts neither the level of UNSAFE play nor the pattern of it, whereas a prompted risk persona moves a checkpoint’s UNSAFE rate by between 50.5 and 98.3 percentage points from its lowest to its highest level (Table 12). An elicited disposition and an instructed persona are therefore not two measurements of one construct; they are different kinds of variable, and the shared axis in Figure 7 aligns them for visual comparison only.

Figure 7 (left) shows how little that null leaves. Human mean UNSAFE play stays inside a 49–62% band across the entire elicitation range, never approaching the low-UNSAFE checkpoints of the pilot roster of §A.12 or its high-UNSAFE Gemini checkpoints. Elicited risk preference does not move a participant far enough to resemble a different checkpoint’s policy.

The right panel is the opposite (Table 12): every checkpoint with a six-level sweep climbs 50.5 to 98.3 points from R1 to R6, all but the two nano checkpoints monotonically, and those two dip by well under a point at R2 before rising. The label does not, however, move checkpoints by a common factor. GPT-5.6 Terra runs lowest at R1 and finishes within 1.3 points of the top, while GPT-5-nano starts fifth of seven and ends last, so the order of the curves is not preserved, and the same instruction is worth very different amounts to different checkpoints. Of the seven tested checkpoints, five have such a sweep; neither Gemini Flash-Lite checkpoint does. In logistic fits that additionally control for the mechanism’s actual private-risk treatment, the coefficients on the agent’s own assigned risk label remain large for the three checkpoints where that controlled fit has been run, so the persona effect is not simply standing in for the mechanical risk treatment.

This contrast does not show that LLM agents possess a latent risk preference comparable to the human elicitation: the human variable is a measured pre-game disposition that turns out to be nearly unrelated to in-game behaviour, whereas the LLM variable is an explicit instruction embedded in the prompt. The more defensible reading is that an assigned risk persona functions as a high-salience behavioural instruction for these models, not as an analogue of a



Figure 6: The persona framings assigned to each company’s executive, and the placebo that separates a persona from the mere presence of an extra sentence. Three families are used. The first assigns no persona at all, and is the neutral condition behind every baseline number in this supplement. The second is a risk-aware family with six ordered levels, R1 to R6, whose wording is adapted from the six-choice Eckel–Grossman gamble scale so that R1 is the most risk-averse framing and R6 the least. The third is a social family that frames the rival either as cooperative or as adversarial. The neutral placebo adds a sentence of matched length that carries no risk content and no social content, so a difference between the placebo and no persona at all is an effect of prompt length rather than of persona. All of these are prompt conditions written into the instruction the model receives. None is a trait measured on the agent, and nothing here is elicited from the model, which is the distinction the rest of this section turns on.

Table 12: LLM risk-persona effect: change in UNSAFE rate from the lowest (R1) to the highest (R6) assigned risk label, and the per-level OLS slope, for every checkpoint with a six-level risk-label sweep ( $n = 360$  players per level,  $n = 330$  for Gemini-3-Flash). The persona coefficient (a logistic fit that additionally controls for the mechanism’s actual private-risk treatment) has been run only for the first three; “not estimated” marks checkpoints for which only the descriptive change and slope are available. Starred rows are outside the nine-route confirmatory baseline and are retained only for exploratory persona analyses.

| Checkpoint      | $\Delta$ UNSAFE (pp) | Per-level slope | Coefficient   |
|-----------------|----------------------|-----------------|---------------|
| GPT-5-nano      | 50.5                 | 0.123           | 0.93          |
| GPT-5.4-nano    | 98.2                 | 0.212           | 1.47          |
| Gemini-3-Flash  | 62.8                 | 0.139           | 2.03          |
| Claude Opus 5   | 81.6                 | 0.152           | not estimated |
| Claude Sonnet 5 | 61.3                 | 0.129           | not estimated |
| GPT-5.6 Luna*   | 71.5                 | 0.156           | not estimated |
| GPT-5.6 Terra*  | 98.3                 | 0.203           | not estimated |

stable human risk disposition. Because the LLM estimates come from a separate risk-matrix experiment, available for five of the seven tested checkpoints and with the private-risk-controlled logistic fit run for only three of those five, this comparison remains exploratory and cross-experimental rather than a like-for-like measurement. The starred rows are outside the nine-route confirmatory baseline and are retained only for exploratory persona analyses.

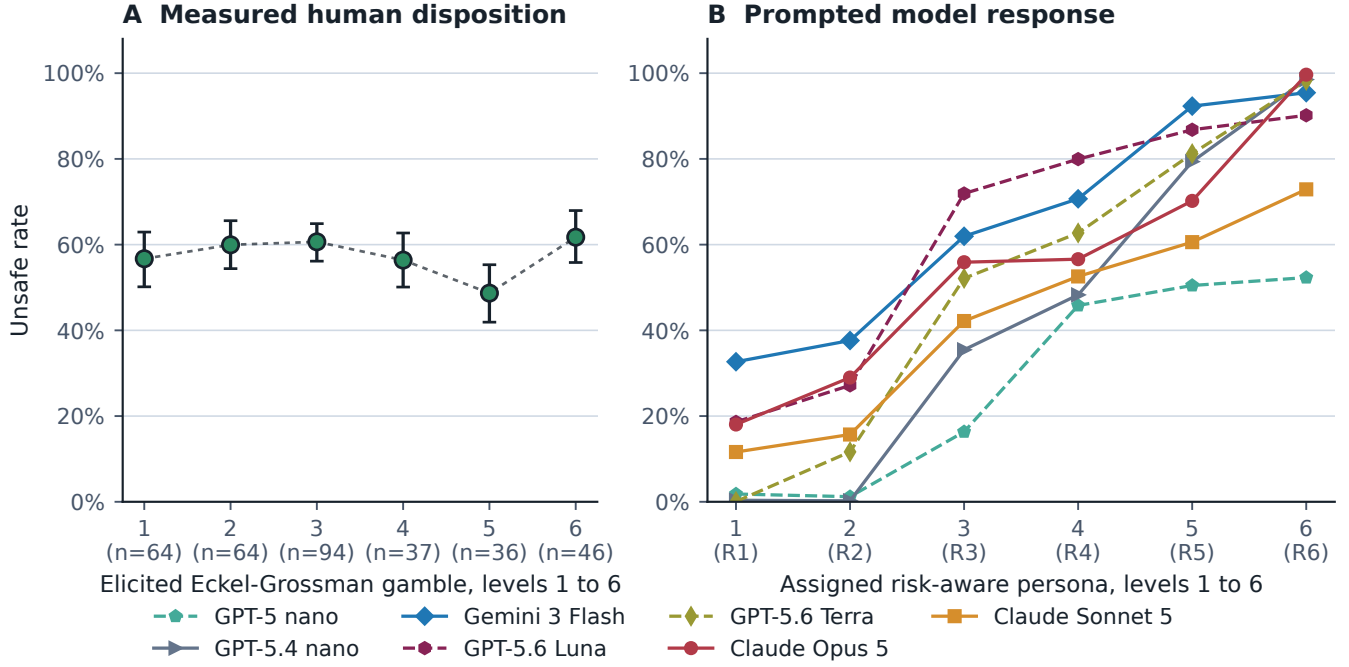
## A.12 The pilot roster behind the projections

The diversity comparison in the main paper reads all nine audited routes on the confirmatory baseline. The projections in this appendix, the behavioural archetypes, the classifier features and the two-dimensional embedding, are built on an earlier export of the

same two-player game covering seven checkpoints. That seven-checkpoint roster is used only for these exploratory analyses; it does not redefine the nine-route human-diversity roster in the main paper. It pools 420 trajectories, 60 per checkpoint, with 340 complete human trajectories from the de-identified dataset of Domingos and Han [5], one incomplete human record excluded,  $N = 760$ . Six of the seven checkpoints carry an endpoint-admission verdict from §A.1; GPT-5 nano is no longer offered on the audited account and so cannot be gated at all, and is retained as description rather than as evidence. Each player is represented by 15 raw features: own action, opponent action, and own-minus-opponent progress gap entering each of rounds 1 to 5, which are the recorded states and actions themselves rather than derived quantities. The archetype projection uses 341 participants rather than 340, because its five descriptor summaries can be computed for one further participant whose complete five-round sequence is unavailable.

No single aggregate rate characterises these checkpoints. Mean UNSAFE play over rounds 1 to 5 is lowest for GPT-5 nano (17%), then Claude Sonnet 5 (22%), Claude Opus 5 (33%), GPT-5.4 nano (54%) and humans (56%), with the Gemini checkpoints running from 73% (Gemini 3 Flash) to 83% (Gemini 3.1 Flash Lite). Closeness to the human mean is, on its own, uninformative about anything beyond that one number, which is why the main paper measures diversity on the action sequences instead.

The archetype projection places each trajectory at the nearest of four behavioural archetypes fitted to the human sample alone (§A.23). The human sample occupies every archetype at every cluster count from two to six, while each checkpoint except GPT-5.4 nano is more concentrated, so that conclusion does not depend on the choice of four (§A.21); the archetypes themselves are exploratory descriptions rather than recovered types, best separated at four groups and only moderately stable under bootstrap. The two-dimensional embedding of the same space is §A.17, and neither



Both panels index the same six ordered levels from 1, and both run from most risk-averse at 1 to least risk-averse at 6: panel A's source file codes those levels 0 to 5, panel B's level  $k$  is persona  $R_k$ . Panel A whiskers are participant-bootstrap 95% intervals; panel B has none, because the persona artifact records a cell mean and a player count, not the player-level rates a bootstrap needs. Panel-B points are means over 360 players (Gemini 3 Flash rests on 90 to 330). Dashed lines mark the 3 routes outside the nine-route audited roster (GPT-5 nano, GPT-5.6 Luna, GPT-5.6 Terra), which carry no admission verdict.

**Figure 7: A measured human disposition against an assigned model label. Left: mean UNSAFE rate by elicited Eckel–Grossman gamble choice, with 95% confidence intervals and participants per bin; the dotted line is the trend across the six group means. Right: mean UNSAFE rate by assigned risk-persona level, for every model with a six-level sweep; GPT-5.6 Luna and Terra (dashed) are the two routes outside the main baseline roster. The shared six-step axis is visual alignment only; the left is a disposition measured once before play, the right an instruction re-assigned every race.**

the main paper nor this appendix rests a conclusion on it: recolouring its identical coordinates by outcome rather than by population separates it almost entirely by UNSAFE rate.

The heterogeneity those pictures suggest is also testable directly, and the test belongs here because it covers five checkpoints rather than the nine the main paper reads. Fitting three nested logistic models to the pooled neutral-baseline pilot, risk only, checkpoint identity only, and checkpoint identity by risk, shows that letting the risk-response profile vary by checkpoint improves fit substantially beyond checkpoint-specific intercepts alone,  $\chi^2(10) = 354.7$ ,  $p < 10^{-3}$ , with several checkpoint-by-risk cells approaching complete separation. A convergence warning in the interaction model means the displayed likelihood-ratio statistic is approximate, so only the conservative threshold is quoted. The fit covers the five-checkpoint pilot baseline only, and the two Claude checkpoints are not in it. Neither caveat changes what it supports: no single risk-response curve represents all five, so the differences between checkpoints are differences in *shape* and not only in level.

### A.13 Population identity in first-five-round trajectories

Table 13 is a supplementary diagnostic: it asks how well a simple classifier can name the population that produced a trajectory, using the same 15 raw features as the trajectory embedding of §A.17. We fit a shallow decision tree with class-balanced loss under stratified five-fold grouped cross-validation. The two seats of an LLM race remain in the same fold, and human trajectories are grouped by source dyad. Since the pooled classes are unequal, balanced accuracy and macro-F1 are the primary measures. A 200-repetition permutation null preserves the observed class counts and the same grouped folds.

Both primary scores exceed chance ( $1/8$ ) and the grouped permutation null (both  $p < 0.001$ ). Raw accuracy is also above its grouped null ( $p = 0.004$ ), but we report it descriptively because the human class is larger. The result supports observable differences in the first-five-round trajectories, while the overlap and the shallow classifier do not support a claim that a latent strategy or model

**Table 13: Dyad/race-grouped population-identity diagnostic on 760 complete first-five-round trajectories. The human population has 340 trajectories and each of the seven listed checkpoints has 60; the classes are therefore imbalanced, so balanced accuracy and macro-F1 are primary. The null interval is the 2.5–97.5 percentile range over grouped label permutations.**

| Metric            | Observed                            | Permutation null mean | Null 95% interval |
|-------------------|-------------------------------------|-----------------------|-------------------|
| Raw accuracy      | $0.282 \pm 0.073$                   | 0.085                 | 0.046–0.171       |
| Balanced accuracy | <b><math>0.384 \pm 0.057</math></b> | 0.124                 | 0.072–0.180       |
| Macro-F1          | <b><math>0.237 \pm 0.051</math></b> | 0.061                 | 0.026–0.108       |

identity is cleanly recoverable. The machine-readable result and its source hashes are released with the analysis code.

#### A.14 Predictive structure of UNSAFE choices

The preceding section shows that populations produce different action patterns. We now ask which recorded parts of the game best predict their next UNSAFE choice. This is a study of association, not cause or internal reasoning. For each population, we fit a random forest, a model that combines many decision trees. Table 14 reports the result and Figure 8 shows the same shares as a heatmap. Predictive scores use five-fold grouped cross-validation, with all decisions from one source dyad or race kept in one fold. The forest predicts decisions after round one from five values known before the choice: the player’s previous action, the opponent’s previous action, the progress gap, the assigned maximum private risk, and the round number. TreeSHAP then measures how much each value contributes to the fitted model’s predictions. A larger SHAP share means that the predictor relied more on that feature. It does not mean that the feature caused the decision.

For human participants, the opponent’s previous action accounts for 56% of total SHAP magnitude, far more than any other feature, while the player’s own previous action contributes only 5%: a reciprocity-dominated profile that independently recovers the pattern found in the human logistic analysis. One caveat has to travel with that reference profile everywhere it is used below, because it is the yardstick the checkpoints are held against and it is the least well identified row in the table. The human forest barely discriminates: ROC AUC 0.62, balanced accuracy 0.57, and a test accuracy of 0.608 against a majority-class baseline of 0.588, so it is about 2.0 points of accuracy above predicting the more common action and nothing else. Adding the demographic covariates does not help, AUC 0.61. The model-side forests are far better fitted, up to AUC 1.00 for Claude Opus 5 and 0.97 for Claude Sonnet 5, so every comparison in this subsection is between shares extracted from very differently identified models. That is not a reason to drop the analysis, and it is a reason not to read a matched ordering as a matched policy. GPT-5-nano is close to the opposite profile: relative race position (progress gap) accounts for 41% of its predictive importance and opponent history for only 7%, and its apparently strong out-of-fold AUC of 0.80 is misleading on its own, since balanced accuracy is exactly 0.50: the checkpoint plays so close to the SAFE floor that the classifier effectively fails to recover the rare

UNSAFE class. Gemini-3-Flash and Gemini-3.1-Flash-Lite are primarily risk-treatment-driven (33% and 37% of SHAP magnitude), consistent with their strong, monotone descriptive risk-response curves. Claude Sonnet 5 is the checkpoint closest to the human profile, and it is the exception to any claim that no model shares it: the opponent’s previous action is its largest predictor too, at 48% against the human 56%. Be exact about how far that goes. Two other checkpoints also rank that feature first, Claude Opus 5 at 42%, which the dagger above marks as collinear with the risk treatment rather than a reciprocity estimate, and Gemini-3.5-Flash-Lite at 33% with its progress gap close behind at 30%. And beyond the leading feature no checkpoint reproduces the human rank order: below the opponent term the human profile runs round number, progress gap, risk, own previous, while Claude Sonnet 5 runs own previous, risk, round number, progress gap. The agreement is in which feature leads, not in the ordering as a whole. GPT-5.4-nano has no comparably dominant predictor: importance is spread across progress gap, risk, round, and its own previous action, with opponent history at 10% and the lowest above-chance predictive fit in the table (AUC 0.54, balanced accuracy 0.52), so its behaviour is comparatively difficult to reconstruct from these five mechanical state variables rather than clearly governed by any one signal.

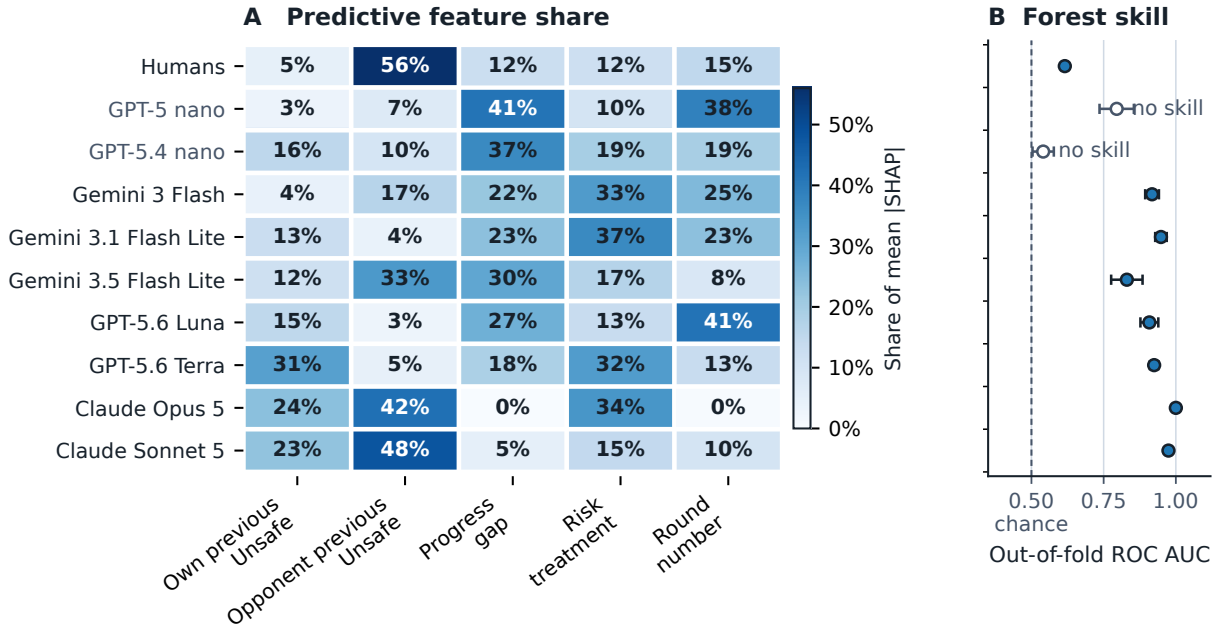
Claude Opus 5 shows why these shares must be read against the design that produced them. Its 42% opponent share reads as strong reciprocity and is not: on the neutral lane it plays UNSAFE on essentially every low-risk decision and essentially none at high risk, so the risk treatment alone separates its choices, and since both seats behave that way the opponent’s previous action is a proxy for the risk level rather than an independent signal. The forest then splits importance between two variables encoding the same event, and the perfect AUC describes a step function, not a better fit. A feature-importance share is interpretable only where the feature keeps variation independent of the others, which is what a near-deterministic policy removes.

One marginal association does agree across populations: human winners play UNSAFE 16.0 percentage points more often than human losers, against an approximately 20-point pooled difference among the checkpoints. The dynamic structure behind that agreement does not agree. GPT-5.4-nano in particular has no single strong predictor, and several checkpoint estimates are weak, reversed in sign, or not reliably estimable because their action distributions sit at the floor or the ceiling.

Predictive performance varies substantially across populations (AUC 0.54 to 1.00), so SHAP shares should not be read as directly comparable measures of decision quality; floor and ceiling behaviour mechanically limits minority-class prediction for several models. The defensible conclusion is narrower than “LLMs use different features”: the same observable state variables organise predictive information differently across populations, and matching a human aggregate action rate does not imply matching the human reciprocity-dominated association profile.

#### A.15 Two-player checkpoint descriptive baseline (full scope)

The five-checkpoint two-player descriptive baseline used by the exploratory observable-driver analysis in this supplement completed



Associational feature use by separately fitted forests; it is not a causal attribution. Panel A normalises every row to 100% whether or not the forest predicts anything, so panel B carries the skill each row rests on: whiskers are the cross-validated standard deviation, and an open marker flags a forest whose balanced accuracy is at chance and whose accuracy does not beat the majority class.

**Figure 8: Population-specific predictive structure for round- $t \geq 2$  UNSAFE choices.** (A) Each row reports the share of mean absolute SHAP value assigned to the same five pre-decision features by a random forest fitted separately to that population. SHAP magnitudes describe how the fitted classifier uses observed features; they are neither causal effects nor evidence of the agents’ internal reasoning. (B) The out-of-fold discrimination each row rests on, because panel A normalises every row to 100% whether or not its forest predicts anything; whiskers are the cross-validated standard deviation and an open marker flags a forest at chance. Read the two panels together, because they are not on the same footing. The human forest discriminates weakly, ROC AUC 0.62 and balanced accuracy 0.57, about 2.0 points of accuracy above guessing the majority class, while the two Claude forests reach 1.00 and 0.97. A share from a weakly fitted forest says mostly where that forest found the little signal there was, so it is a much looser proxy for the player’s policy than a share from a well fitted one.

**Table 14: Population-specific random-forest prediction of UNSAFE play.** Feature columns report shares of total mean absolute SHAP magnitude (rows sum to 100%, up to rounding); AUC and balanced accuracy are mean out-of-fold values from the population-specific race/dyad-grouped five-fold split.

| Population            | Own previous | Opponent previous | Progress gap | Risk | Round | OOF AUC | Balanced accuracy |
|-----------------------|--------------|-------------------|--------------|------|-------|---------|-------------------|
| Human                 | 5%           | 56%               | 12%          | 12%  | 15%   | 0.62    | 0.57              |
| GPT-5-nano            | 3%           | 7%                | 41%          | 10%  | 38%   | 0.80    | 0.50              |
| GPT-5.4-nano          | 16%          | 10%               | 37%          | 19%  | 19%   | 0.54    | 0.52              |
| Gemini-3-Flash        | 4%           | 17%               | 22%          | 33%  | 25%   | 0.92    | 0.77              |
| Gemini-3.1-Flash-Lite | 13%          | 4%                | 23%          | 37%  | 23%   | 0.95    | 0.75              |
| Gemini-3.5-Flash-Lite | 12%          | 33%               | 30%          | 17%  | 8%    | 0.83    | 0.71              |
| GPT-5.6-Luna          | 15%          | 3%                | 27%          | 13%  | 41%   | 0.91    | 0.78              |
| GPT-5.6-Terra         | 31%          | 5%                | 18%          | 32%  | 13%   | 0.92    | 0.85              |
| Claude Opus 5         | 24%          | 42% <sup>†</sup>  | 0%           | 34%  | 0%    | 1.00    | 1.00              |
| Claude Sonnet 5       | 23%          | 48%               | 5%           | 15%  | 10%   | 0.97    | 0.93              |

<sup>†</sup> Collinear with the risk treatment, not a reciprocity estimate; see the discussion below.

150 races and 2,790 decisions, crossing five checkpoints (GPT-5-nano, GPT-5.4-nano, Gemini-3-Flash-preview, Gemini-3.1-Flash-Lite, Gemini-3.5-Flash-Lite) with the mechanism’s three assigned private-risk levels (0.10, 0.60, 0.90). Read as descriptive curves rather

than as pooled rates, GPT-5-nano’s UNSAFE rate was low and nearly flat across the three risk levels; GPT-5.4-nano’s response to risk level was non-monotone; and the three Gemini checkpoints’ UNSAFE rate

declined as the assigned maximum risk increased, the opposite direction from the two OpenAI checkpoints. Because provider routes, protocol signatures, and pilot sample sizes differ across checkpoints, these five curves demonstrate qualitative heterogeneity in how each checkpoint responds to the risk manipulation; we do not pool them into a single model-family estimate, and the formal test of that heterogeneity is the checkpoint-by-risk interaction of §A.12. Table 15 gives the per-checkpoint, per-risk-level rates underlying the qualitative curve descriptions above.

**Table 15: Player-level UNSAFE rate by checkpoint and assigned maximum-private-risk level, five-checkpoint two-player descriptive baseline ( $n = 20$  players per checkpoint-risk cell, ten races each).**

| Checkpoint            | Risk 0.10 | Risk 0.60 | Risk 0.90 |
|-----------------------|-----------|-----------|-----------|
| GPT-5 nano            | 12.3%     | 15.1%     | 14.5%     |
| GPT-5.4 nano          | 57.7%     | 49.6%     | 56.7%     |
| Gemini 3 Flash        | 100.0%    | 72.3%     | 53.9%     |
| Gemini 3.1 Flash Lite | 100.0%    | 80.1%     | 69.9%     |
| Gemini 3.5 Flash Lite | 83.8%     | 70.8%     | 62.6%     |

#### E. Exploratory interaction scope.

#### A.16 $N$ -player self-play scope ( $N = 3, 4, 5$ )

*The multiplayer payoff rule.* With more than two companies there is no single opponent to index the two-player matrix against, so the stage payoff follows the group-count rule of Han et al. [7]. In a round where  $k$  of the  $N$  companies choose SAFE, a company that chose SAFE receives  $\pi_S(k)$  and a company that chose UNSAFE receives  $\pi_U(k)$ :

$$\pi_S(k) = \begin{cases} -c + \frac{b}{N}, & k = N, \\ -c + \frac{b}{k + s(N - k)}, & 1 \leq k < N, \end{cases} \quad (1)$$

$$\pi_U(k) = \frac{sb}{k + s(N - k)}, \quad 0 \leq k < N.$$

Here  $b = 4$  is the benefit shared among the companies in each round,  $c = 1$  is the cost of playing SAFE in a round, and  $s = 1.5$  is the speed of UNSAFE development relative to SAFE, which is also the weight of an UNSAFE company’s share. The source rule also lets an UNSAFE company be found out with some probability; that term is set to zero here. At  $N = 2$  the rule gives the two-player stage payoffs of the main paper exactly: 1.0, 0.6, 2.4 and 2.0. Everything else carries over unchanged. SAFE adds 1.0 to progress and UNSAFE adds 1.5. The hidden horizon lasts at least five rounds and stops with probability 0.2 after each round from the fifth on. The prize  $B = 100$  is split evenly among the companies with the highest progress, and the private risk rule and its three levels are the same.

*Group-size scope.* The exploratory multiplayer position analysis reports position effects for GPT-5-nano and GPT-5.4-nano at fixed  $N$ ; this subsection gives the broader group-size scope. Matched OpenAI self-play baselines cover  $N \in \{3, 4, 5\}$  for both checkpoints: 180 pilot races, 720 player trajectories, and 6,120 decisions in total,

pooled across the mechanism’s three risk levels. Table 16 reports the aggregate UNSAFE rate by checkpoint and group size.

**Table 16: Aggregate UNSAFE rate by group size, OpenAI self-play, pooled across the three assigned private-risk levels. Ten independent races per model-risk- $N$  cell.**

| Checkpoint   | $N = 3$ | $N = 4$ | $N = 5$ |
|--------------|---------|---------|---------|
| GPT-5-nano   | 16.2%   | 26.8%   | 21.9%   |
| GPT-5.4-nano | 83.7%   | 87.0%   | 84.0%   |

Neither checkpoint’s response to group size is monotone in  $N$ , and the association is checkpoint-specific: GPT-5-nano’s rate roughly doubles from  $N = 3$  to  $N = 4$  before partially receding at  $N = 5$ , while GPT-5.4-nano stays within a narrow 83–87% band throughout. We do not read either pattern as an isolated group-size effect, for two reasons. First, changing  $N$  also changes the stage-payoff rule itself (Equation 1 uses the count  $k$  of SAFE-choosing companies rather than a single opponent’s move), the number of opponents, and prompt length, so  $N$  is confounded with the representation of the game as well as with its size. Second, the pilots use only ten independent races per model-risk- $N$  cell, which is a small base for the per-cell rates entering Table 16 and leaves individual cells vulnerable to the same kind of small- $n$  noise flagged for specific rank  $\times$  risk-band cells in the exploratory position analysis. Table 17 disaggregates Table 16 by assigned risk level and reports a 95% percentile bootstrap interval over the ten independent races in each cell. The decision count is retained only as a descriptive row count.

**Table 17: UNSAFE rate by checkpoint, group size, and assigned maximum-private-risk level, OpenAI  $N$ -player self-play. Intervals are 95% percentile bootstrap intervals over race clusters, and  $n_{\text{race}}$  is the independent-unit count.**

| $N$ | Checkpoint   | Risk | $n_{\text{race}}$ | UNSAFE rate (95% CI) |
|-----|--------------|------|-------------------|----------------------|
| 3   | GPT-5 nano   | 0.10 | 10                | 17.4% (14.5–20.0%)   |
| 3   | GPT-5 nano   | 0.60 | 10                | 18.1% (13.3–22.4%)   |
| 3   | GPT-5 nano   | 0.90 | 10                | 11.3% (6.1–16.9%)    |
| 3   | GPT-5.4 nano | 0.10 | 10                | 85.4% (82.5–88.5%)   |
| 3   | GPT-5.4 nano | 0.60 | 10                | 83.7% (78.5–88.7%)   |
| 3   | GPT-5.4 nano | 0.90 | 10                | 83.3% (80.0–86.9%)   |
| 4   | GPT-5 nano   | 0.10 | 10                | 19.2% (14.5–23.8%)   |
| 4   | GPT-5 nano   | 0.60 | 10                | 27.8% (21.6–33.3%)   |
| 4   | GPT-5 nano   | 0.90 | 10                | 24.1% (15.5–32.1%)   |
| 4   | GPT-5.4 nano | 0.10 | 10                | 88.1% (85.9–90.3%)   |
| 4   | GPT-5.4 nano | 0.60 | 10                | 88.4% (86.2–90.6%)   |
| 4   | GPT-5.4 nano | 0.90 | 10                | 83.8% (80.3–87.5%)   |
| 5   | GPT-5 nano   | 0.10 | 10                | 18.9% (14.7–22.4%)   |
| 5   | GPT-5 nano   | 0.60 | 10                | 20.0% (16.1–23.6%)   |
| 5   | GPT-5 nano   | 0.90 | 10                | 22.7% (18.7–26.9%)   |
| 5   | GPT-5.4 nano | 0.10 | 10                | 80.0% (75.4–84.9%)   |
| 5   | GPT-5.4 nano | 0.60 | 10                | 87.1% (84.5–89.6%)   |
| 5   | GPT-5.4 nano | 0.90 | 10                | 85.1% (82.5–88.0%)   |

### A.17 Trajectory embedding recoloured by outcome

The embedding of raw five-round trajectories separates almost entirely by how much UNSAFE play a trajectory carries. Both colourings of it live here. Figure 9 colours it by population, and Figure 10 is the check: the same coordinates, with only the colouring changed, so any structure visible in the second is structure the embedding already had. Reading them in that order is the point. The population panels look like eight different regions; recolouring the identical points by each player’s own UNSAFE rate splits the embedding into a SAFE region, an UNSAFE region and a thin mixed band, with very few points on the wrong side. The apparent separation of populations is therefore largely a separation of action rates, which is why the main paper rests no conclusion on either picture and uses the grouped classifier of Table 13 instead.

### A.18 Mean round-by-round profile per population

The trajectory-diversity comparison in the main paper shows that human trajectories spread more widely than any single check-point’s. Figure 11 is the same comparison read one feature at a time: population means over the fourteen features every trajectory shares, with action probabilities and absolute progress gaps separated so that neither scale obscures the other.

### A.19 Player-level UNSAFE rate by population

Figure 12 is the aggregate level of the same comparison: one number per population, the mean share of a player’s first five decisions that were UNSAFE. It is the plainest form of the point that a matching aggregate rate does not imply matching behaviour, because a population can reach a middling rate either by mixing or by every player behaving the same way.

### A.20 UNSAFE rate by persona band and relative position

The main paper does not report these cells. Tables 18 and 19 give them in full for the two checkpoints that have a persona sweep and an observed position. Relative position is observed rather than assigned, so these are associations within a persona band, not position effects. The race is this paper’s independent unit. Race-level identifiers were not retained in the archived derivative used for this breakdown; we therefore report these cells descriptively and make no inferential claim from them. We print only the decision count and descriptive rate, with no confidence interval.

### A.21 Cluster-count sensitivity and stability of the human archetypes

The behavioural archetypes used in this supplement are  $k$ -means groups: the 341 human participants are described by five standardised summary numbers each, and  $k$ -means partitions them into  $k$  groups so that each participant is near the average of its own group. The number of groups is a choice, not a measurement, so this subsection reports what changes when that choice changes. The five descriptors are the overall UNSAFE rate, reciprocity, position sensitivity, own-action autocorrelation, and whether the first round

**Table 18: UNSAFE rate by persona band and relative position, GPT-5-nano.  $n$  is the number of decisions in the cell. These cells are descriptive only and support no inference.**

| Persona band | Position | $n$  | UNSAFE rate |
|--------------|----------|------|-------------|
| Baseline     | Leader   | 1460 | 19.7%       |
| Baseline     | Middle   | 1065 | 30.4%       |
| Baseline     | Trailer  | 175  | 33.1%       |
| Low (R1–R2)  | Leader   | 1094 | 5.9%        |
| Low (R1–R2)  | Middle   | 198  | 14.6%       |
| Low (R1–R2)  | Trailer  | 58   | 12.1%       |
| Mid (R3–R4)  | Leader   | 649  | 40.7%       |
| Mid (R3–R4)  | Middle   | 451  | 38.1%       |
| Mid (R3–R4)  | Trailer  | 250  | 32.8%       |
| High (R5–R6) | Leader   | 702  | 63.2%       |
| High (R5–R6) | Middle   | 339  | 54.3%       |
| High (R5–R6) | Trailer  | 309  | 50.2%       |

**Table 19: UNSAFE rate by persona band and relative position, GPT-5.4-nano.  $n$  is the number of decisions in the cell and cells marked  $^\dagger$  have  $n < 20$ . These cells are descriptive only and support no inference.**

| Persona band | Position | $n$           | UNSAFE rate |
|--------------|----------|---------------|-------------|
| Baseline     | Leader   | 1649          | 83.6%       |
| Baseline     | Middle   | 703           | 84.8%       |
| Baseline     | Trailer  | 348           | 86.5%       |
| Low (R1–R2)  | Leader   | 1073          | 3.3%        |
| Low (R1–R2)  | Middle   | 268           | 1.5%        |
| Low (R1–R2)  | Trailer  | 9 $^\dagger$  | 0.0%        |
| Mid (R3–R4)  | Leader   | 1164          | 95.7%       |
| Mid (R3–R4)  | Middle   | 70            | 88.6%       |
| Mid (R3–R4)  | Trailer  | 116           | 91.4%       |
| High (R5–R6) | Leader   | 1328          | 99.3%       |
| High (R5–R6) | Middle   | 4 $^\dagger$  | 100.0%      |
| High (R5–R6) | Trailer  | 18 $^\dagger$ | 100.0%      |

was UNSAFE. They are standardised on the human sample alone, and every other population is later projected into that same human space rather than re-standardised.

Table 20 gives the diagnostics for  $k = 2$  to 6. Three of them describe how tight and separated the groups are: the silhouette coefficient, which is higher when a participant is closer to its own group than to the nearest other one; the Calinski–Harabasz index, higher when between-group spread is large relative to within-group spread; and the Davies–Bouldin index, which is *lower* when groups are better separated. The fourth, inertia, is the total squared distance to group centres and falls mechanically as  $k$  grows, so it is reported for completeness rather than as evidence. The choice of  $k = 4$  used elsewhere is the best cell on two of the three interpretable indices, with the highest silhouette (0.316, against 0.278 at  $k = 2$  and 0.241 at  $k = 5$ ) and the lowest Davies–Bouldin (1.218).

Stability is only moderate, and we say so rather than presenting the groups as recovered types. The bootstrap adjusted Rand index falls from 0.961 at  $k = 2$ , where the split is nearly always the same one, to 0.717 at  $k = 3$ , 0.626 at  $k = 4$ , 0.458 at  $k = 5$ , and 0.518

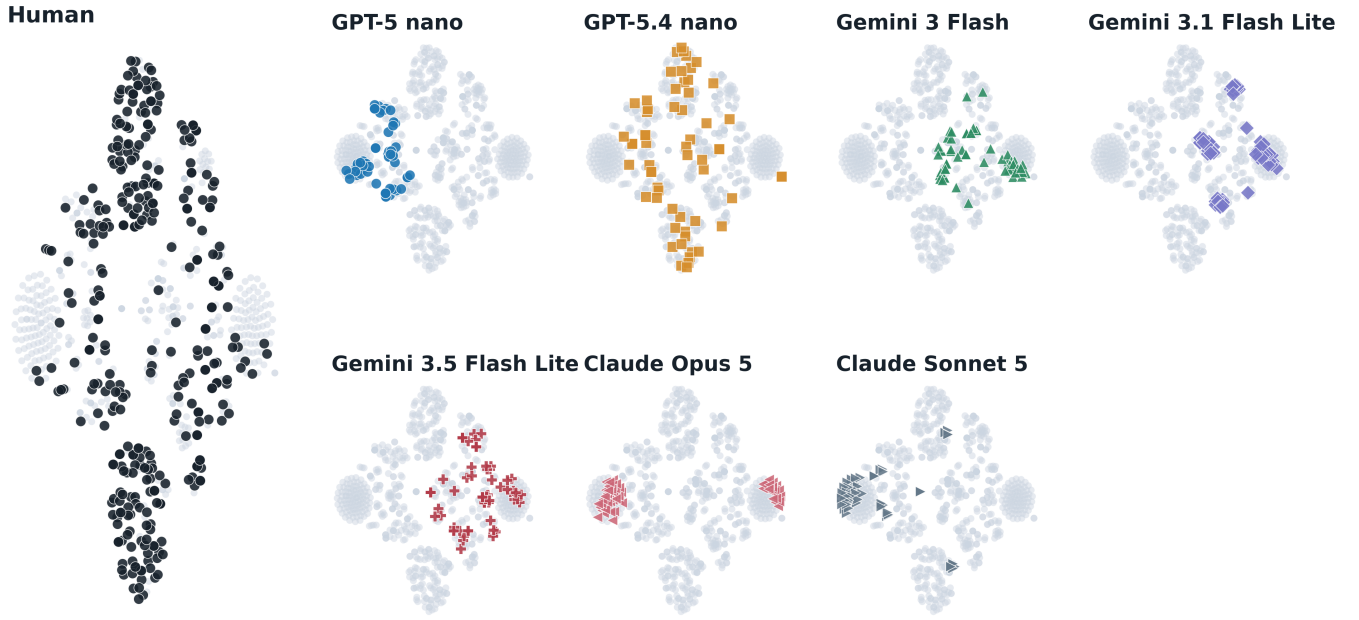


Figure 9: The population-coloured embedding, shown here so that Figure 10 can be compared against it directly. A t-SNE embedding places each of the 760 complete first-five-round trajectories at a point in two dimensions, arranged so that trajectories with similar action-and-position histories fall near each other; the input is the pooled 15-dimensional first-five-round action-and-position space, the same input as the population-identity classifier below. One panel is drawn per population, human first. The axes have no units and no direction: only relative closeness carries meaning, and distances between well-separated clusters are not interpretable. The main paper rests no conclusion on this figure, and the reason is Figure 10: recolouring these identical coordinates by each player’s own UNSAFE rate separates the embedding almost entirely, so what looks like a separation of populations here is largely a separation of action rates. The quantitative statement about population identity is the grouped classifier of Table 13, not this picture.

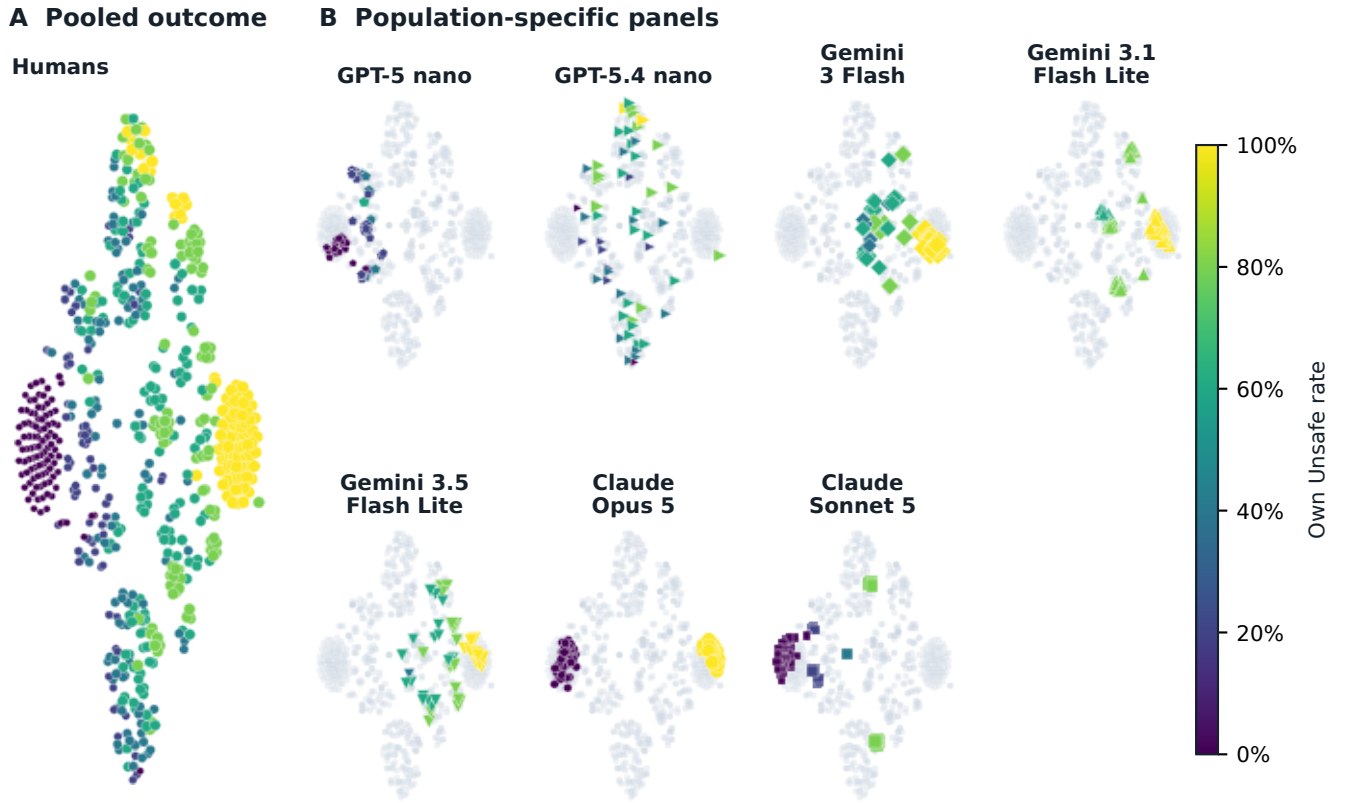
Table 20: Cluster-count sweep for the human behavioural archetypes,  $k = 2$  to 6, on the same matrix of 341 human participants by five standardised descriptors, seed 0. Higher is better for silhouette and Calinski–Harabasz, lower is better for Davies–Bouldin; inertia falls with  $k$  by construction. “Group sizes” is the number of participants per group. The adjusted Rand index (ARI) measures how much a partition refitted on a resampled sample agrees with the partition fitted on the full sample: 1 is identical, 0 is chance agreement. It is computed over 500 bootstrap resamples of participants, and the interval is the 2.5 to 97.5 percentile range across those resamples.

| $k$ | Silhouette | Calinski<br>Harabasz | Davies<br>Bouldin | Inertia | Group sizes                 | Smallest<br>group $n$ | ARI mean | ARI 95% interval |
|-----|------------|----------------------|-------------------|---------|-----------------------------|-----------------------|----------|------------------|
| 2   | 0.278      | 110.9                | 1.578             | 1280.9  | 187 / 154                   | 154                   | 0.961    | 0.76–1.00        |
| 3   | 0.267      | 91.7                 | 1.542             | 1102.1  | 107 / 63 / 171              | 63                    | 0.717    | 0.37–0.98        |
| 4   | 0.316      | 83.5                 | 1.218             | 975.3   | 19 / 170 / 133 / 19         | 19                    | 0.626    | 0.39–0.98        |
| 5   | 0.241      | 78.2                 | 1.363             | 880.2   | 20 / 57 / 118 / 87 / 59     | 20                    | 0.458    | 0.30–0.69        |
| 6   | 0.239      | 79.0                 | 1.405             | 780.2   | 42 / 78 / 56 / 55 / 91 / 19 | 19                    | 0.518    | 0.34–0.79        |

at  $k = 6$ . At  $k = 4$  the interval runs from 0.39 to 0.98, which is wide. The groups are therefore useful descriptions of where the human sample sits in this five-number space, not evidence that four discrete human strategies exist.

The supplementary coverage result does not depend on  $k$ . It concerns how the human sample spreads across the whole set of archetypes while almost every single checkpoint concentrates on a few. Table 21 counts, for each population and each  $k$ , how many

of the  $k$  archetypes that population actually occupies, where a population counts as occupying an archetype when at least 5% of its entities project to it. The human row occupies all  $k$  archetypes at every  $k$  from two to six. No checkpoint matches that. The closest is the exception this supplement names, GPT-5.4 nano, which occupies all  $k$  archetypes at  $k = 2, 3, 5$  and 6 and three of four at  $k = 4$ . Every other checkpoint falls short from  $k = 4$  upward, and GPT-5 nano is the extreme case, occupying exactly one archetype at every  $k$ .



**Figure 10: The same t-SNE coordinates as Figure 9, recoloured by each player's own mean UNSAFE rate over rounds 1–5 on a monotonic viridis low-to-high rate scale instead of population identity. Left: all 760 trajectories pooled. Right: the same colouring broken out per population.**

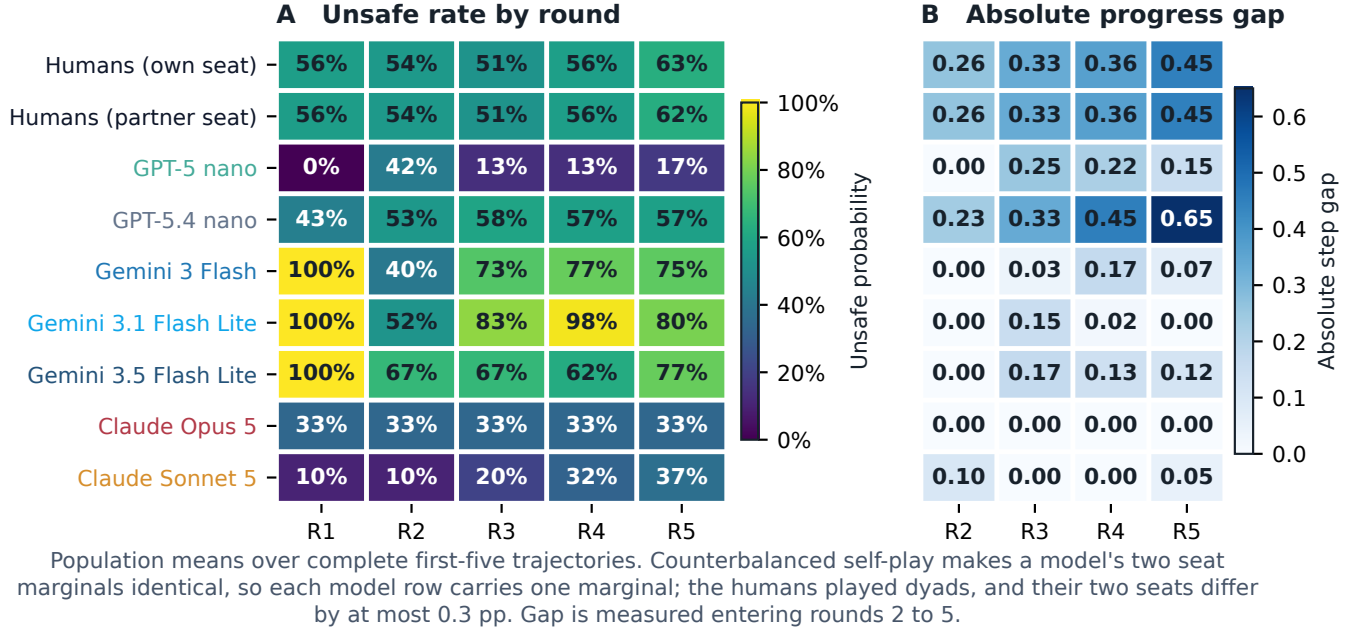
The 5% threshold is a reporting choice, and the artifact records the counts at a 1% threshold and at strictly positive occupancy alongside it so the claim can be read at any of the three.

One property of these descriptors is ours to disclose rather than a reviewer's to discover. Two of the five are undefined for some participants, and the clustering fills the gap with the sample mean instead of dropping the participant, exactly as the published clustering does. Position sensitivity, which contrasts a participant's behaviour when ahead against when behind, is undefined for 268 of the 341 participants, because those participants were never observed both ahead and behind; it is mean-imputed for all 268, that is for 79% of the sample. Reciprocity and own-action autocorrelation are each undefined and mean-imputed for 38 participants. The overall UNSAFE rate and the first-round action are complete for all 341. A mean-imputed cell carries no information about the participant it is attached to, so position sensitivity does most of its work on under a quarter of the sample, and the archetype signatures should be read with that in mind. No participant is excluded at any step: all 341 receive a cluster label.

## A.22 Trajectory-diversity statistics

The main paper reports that human first-five-round trajectories are more varied than most admitted route-risk cells under a cluster-aware null. The primary dyad-clustered summary is in Table 22. Table 24 retains the full nine-route table, with participant-level human resampling and the companion Hill and Hamming statistics. Together they make each value checkable against its population, its risk condition, and the independent unit it rests on. Table 23 is the earlier seven-checkpoint pilot export, kept because it carries GPT-5 nano, an endpoint no longer offered on the audited identity and therefore holding no admission verdict. That export has no row for GPT-5.4 or GPT-5.5, which are two of the five admitted routes, so it cannot settle a claim about the admitted set and is not used for one.

A trajectory here is the pair of actions taken by the two players over the first five rounds, that is ten SAFE or UNSAFE bits. Three statistics describe how varied a set of such trajectories is. Hill  $q = 0$  counts how many *distinct* trajectories appear, so it treats a trajectory seen once and a trajectory seen forty times alike. Hill  $q = 1$  is the effective number of trajectories, the exponential of Shannon entropy, which discounts trajectories that are rare: it equals  $q = 0$  when every trajectory is equally common and falls below it when



**Figure 11: Round-by-round profile for each population, shown as two source-backed heatmaps. Panel A reports the probability of an UNSAFE action for each player's own and opponent action at rounds 1–5. Panel B reports the mean absolute progress gap entering rounds 2–5, so it measures how far apart the two players are after pooling focal-player roles. The eight rows are the human reference and the seven baseline model populations.**

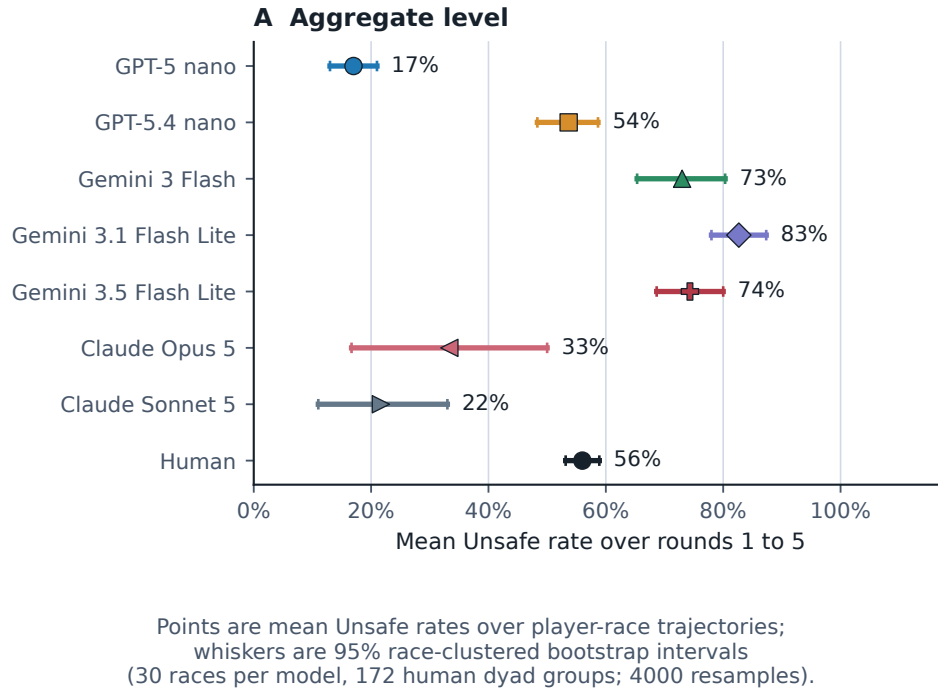
**Table 21: Archetype occupancy by population and cluster count. Each cell counts how many of the  $k$  human-fitted archetypes that population occupies, where occupying means that at least 5% of the population's entities project nearest to that archetype.  $n$  is the number of entities projected: participants for the human row, no-persona player-races for each checkpoint. A row that reaches  $k$  at every  $k$  covers the full human-fitted space; the human reference does so at every  $k$  and no checkpoint does.**

| Population            | $n$ | $k=2$ | $k=3$ | $k=4$ | $k=5$ | $k=6$ |
|-----------------------|-----|-------|-------|-------|-------|-------|
| Human (reference)     | 341 | 2     | 3     | 4     | 5     | 6     |
| GPT-5 nano            | 120 | 1     | 1     | 1     | 1     | 1     |
| GPT-5.4 nano          | 120 | 2     | 3     | 3     | 5     | 6     |
| Gemini 3 Flash        | 120 | 1     | 1     | 2     | 3     | 3     |
| Gemini 3.1 Flash Lite | 60  | 1     | 1     | 2     | 2     | 3     |
| Gemini 3.5 Flash Lite | 60  | 1     | 1     | 2     | 3     | 3     |
| GPT-5.6 Luna          | 120 | 2     | 1     | 2     | 2     | 3     |
| GPT-5.6 Terra         | 120 | 2     | 2     | 3     | 3     | 4     |
| Claude Opus 5         | 120 | 2     | 2     | 2     | 2     | 2     |
| Claude Sonnet 5       | 120 | 2     | 3     | 2     | 3     | 4     |

a few dominate. Mean pairwise Hamming distance is the average number of the ten bits on which two trajectories drawn from the set differ, divided by ten, so 0 means every trajectory in the set is identical and 0.5 means two trajectories differ on half their bits on average.

Sample sizes differ across populations, and both counting statistics rise with sample size, so every cell is compared at the same size. The point estimate is the mean over draws of 20 trajectories taken without replacement from the cell, or the exact cell value where the cell already holds exactly 20. The interval is a two-stage cluster bootstrap: clusters are resampled with replacement first, and only then is the draw rarefied to 20 inside each resample, over 2,000 resamples. The cluster is the independent unit of the design: a source dyad for the human rows, with singleton complete trajectories retained, and a race for the checkpoint rows, because the two seats of a race share the same sampled horizon and setback draws and so are not independent of each other. The  $n_{\text{cluster}}$  column is that count. The human-side  $q=0$  comparison used for the main figure is instead cluster-aware at the dyad level: it samples ten complete human dyads, retaining twenty trajectories but ten interaction units. Four complete human trajectories have no complete partner in the de-identified file and are omitted from that primary dyad null.

The ordering in Table 23 is consistent across all three risk levels and all three statistics. The human rows sit near the ceiling of the comparison: 19.2, 19.7, and 19.5 distinct trajectories out of a possible 20, with mean pairwise Hamming distance between 0.467 and 0.497, close to the 0.5 that two unrelated trajectories would give. Only one checkpoint in this pilot export reaches that: GPT-5.4 nano returns 19.0 distinct trajectories at every risk level, with Hamming between 0.478 and 0.506. The rest are far below. Claude Opus 5 returns exactly one distinct trajectory at all three risk levels, with Hamming 0, meaning every one of its player-races produced the identical ten-action sequence. Gemini 3 Flash and Gemini 3.1 Flash



**Figure 12: Player-level mean UNSAFE rate over rounds 1 to 5, one row per population. The point is the unweighted mean over player-race trajectories. The whisker is a 95% percentile bootstrap interval that resamples *clusters*, not single trajectories, because the two seats of a race are not independent of each other: they share the sampled stopping horizon and the setback draws. The cluster is therefore the race for every checkpoint row and the source dyad for the human row, with singleton complete trajectories retained as singleton clusters; the interval uses 4,000 resamples. The point estimates are unchanged from a trajectory-level calculation; only the interval changed, and it supersedes an earlier interval that resampled player-race trajectories and so treated the two seats of a race as independent.**

**Table 22: Primary dyad-clustered human null for distinct paired first-five-round trajectories. Each draw samples ten complete dyads, contributing twenty trajectories, without replacement from the risk pool. The mean and 95% RI are over 20,000 draws; the minimum is the finite-draw floor used for the strict below-null comparison. “Admitted below” counts the five screened routes whose confirmatory  $q=0$  value is below every draw; “all routes below” uses all nine audited routes.**

| Risk | Dyads | Mean | 95% RI | Min | Admitted | All routes |
|------|-------|------|--------|-----|----------|------------|
| 0.1  | 48    | 18.6 | 16–20  | 12  | 5/5      | 7/9        |
| 0.6  | 52    | 19.6 | 18–20  | 15  | 5/5      | 7/9        |
| 0.9  | 68    | 19.2 | 17–20  | 13  | 3/5      | 4/9        |

Lite collapse the same way at risk 0.1. Claude Sonnet 5 falls from 5.0 distinct trajectories at risk 0.1 to 3.0 at risk 0.9, with Hamming 0.020 at the latter. This is the shape of the finding: a checkpoint can match a human aggregate action rate while producing a small handful of distinct action sequences, or only one.

Table 24 is the version of that sensitivity that can be read against the admission gate, because it has a row for every route the gate

ruled on. The primary  $q=0$  model entries in the table, read against the dyad floors in Table 22, put 13 of 15 admitted cells below every dyad draw. The two admitted boundary cells are GPT-5.4 and GPT-5.5 at risk 0.9; both have 13 or 14 distinct trajectories against a dyad floor of 13. The participant-level Hill  $q=1$  sensitivity places every admitted route below the human interval at all three risk levels. GPT-5.4 mini and GPT-5.4 nano remain the two refused routes that reach the participant-level human range at every risk. Claude Opus 5 again returns exactly one distinct trajectory at all three risk levels, with Hamming 0. The dyad correction changes the scope of the  $q=0$  claim, not the concentration result itself.

### A.23 Behavioural-archetype coverage by population

This supplement reports that human trajectories spread more evenly across the archetypes than most single checkpoints do, with GPT-5.4-nano the exception. Table 25 is the per-population breakdown behind that statement at the four archetypes used throughout, so a single share can be checked against its population and its archetype; Table 21 is the same comparison read as a count, at every cluster count from two to six.

**Table 23: Participant-level sensitivity for sample-size-matched diversity of paired first-five-round trajectories, by maximum-private-risk level and population. A trajectory is the two players’ ten actions over rounds 1 to 5. Hill  $q=0$  is the number of distinct trajectories, Hill  $q=1$  the effective number after discounting rare ones, and Hamming the mean fraction of the ten bits on which two trajectories differ.  $n_{\text{src}}$  is the trajectories available in the cell and  $n_{\text{cluster}}$  the independent units it rests on: participants for the human rows, races for the checkpoint rows. Every cell is compared at 20 trajectories. The parenthesised ranges are 95% *resampling intervals* (RI) and not confidence intervals: they are two-stage cluster bootstrap ranges, with clusters resampled with replacement and then rarefied to 20 within each resample, over 2,000 resamples, and they bound how much the statistic moves under resampling rather than making a confidence statement about the observed count. Because the point estimate is the observed cell value while the interval is computed on resampled clusters, which lose distinct trajectories to duplication, an interval can sit below its point estimate.**

| Risk | Population            | $n_{\text{src}}$ | Hill $q=0$ (95% RI) | Hill $q=1$ (95% RI) | Hamming (95% RI)    | $n_{\text{cluster}}$ |
|------|-----------------------|------------------|---------------------|---------------------|---------------------|----------------------|
| 0.1  | Human                 | 98               | 19.2 (15.0–20.0)    | 18.9 (13.2–20.0)    | 0.467 (0.407–0.504) | 98                   |
|      | GPT-5 nano            | 20               | 13.0 (5.0–12.0)     | 10.2 (3.4–10.7)     | 0.275 (0.163–0.339) | 10                   |
|      | GPT-5.4 nano          | 20               | 19.0 (9.0–16.0)     | 18.7 (7.3–15.2)     | 0.478 (0.405–0.505) | 10                   |
|      | Gemini 3 Flash        | 20               | 1.0 (1.0–1.0)       | 1.0 (1.0–1.0)       | 0.000 (0.000–0.000) | 10                   |
|      | Gemini 3.1 Flash Lite | 20               | 1.0 (1.0–1.0)       | 1.0 (1.0–1.0)       | 0.000 (0.000–0.000) | 10                   |
|      | Gemini 3.5 Flash Lite | 20               | 5.0 (3.0–5.0)       | 3.6 (1.9–4.8)       | 0.166 (0.076–0.227) | 10                   |
|      | Claude Opus 5         | 20               | 1.0 (1.0–1.0)       | 1.0 (1.0–1.0)       | 0.000 (0.000–0.000) | 10                   |
|      | Claude Sonnet 5       | 20               | 5.0 (3.0–5.0)       | 4.5 (2.6–5.0)       | 0.354 (0.246–0.413) | 10                   |
| 0.6  | Human                 | 104              | 19.7 (15.0–20.0)    | 19.7 (14.1–20.0)    | 0.495 (0.455–0.513) | 104                  |
|      | GPT-5 nano            | 20               | 8.0 (5.0–8.0)       | 7.2 (3.4–7.7)       | 0.244 (0.154–0.301) | 10                   |
|      | GPT-5.4 nano          | 20               | 19.0 (9.0–16.0)     | 18.7 (7.2–15.2)     | 0.506 (0.402–0.520) | 10                   |
|      | Gemini 3 Flash        | 20               | 11.0 (5.0–11.0)     | 9.3 (4.4–10.2)      | 0.312 (0.208–0.371) | 10                   |
|      | Gemini 3.1 Flash Lite | 20               | 4.0 (2.0–4.0)       | 2.8 (2.0–3.7)       | 0.229 (0.211–0.255) | 10                   |
|      | Gemini 3.5 Flash Lite | 20               | 12.0 (6.0–12.0)     | 11.5 (5.5–10.9)     | 0.364 (0.267–0.397) | 10                   |
|      | Claude Opus 5         | 20               | 1.0 (1.0–1.0)       | 1.0 (1.0–1.0)       | 0.000 (0.000–0.000) | 10                   |
|      | Claude Sonnet 5       | 20               | 5.0 (1.0–5.0)       | 2.7 (1.0–4.0)       | 0.074 (0.000–0.133) | 10                   |
| 0.9  | Human                 | 138              | 19.5 (16.0–20.0)    | 19.3 (14.8–20.0)    | 0.497 (0.457–0.516) | 138                  |
|      | GPT-5 nano            | 20               | 12.0 (6.0–12.0)     | 10.0 (3.9–10.7)     | 0.232 (0.134–0.304) | 10                   |
|      | GPT-5.4 nano          | 20               | 19.0 (9.0–16.0)     | 18.7 (7.3–15.2)     | 0.486 (0.394–0.504) | 10                   |
|      | Gemini 3 Flash        | 20               | 11.0 (6.0–11.0)     | 10.7 (5.5–10.2)     | 0.311 (0.262–0.314) | 10                   |
|      | Gemini 3.1 Flash Lite | 20               | 3.0 (2.0–3.0)       | 2.9 (2.0–3.0)       | 0.101 (0.079–0.105) | 10                   |
|      | Gemini 3.5 Flash Lite | 20               | 12.0 (6.0–12.0)     | 11.5 (5.2–10.9)     | 0.401 (0.319–0.417) | 10                   |
|      | Claude Opus 5         | 20               | 1.0 (1.0–1.0)       | 1.0 (1.0–1.0)       | 0.000 (0.000–0.000) | 10                   |
|      | Claude Sonnet 5       | 20               | 3.0 (1.0–3.0)       | 1.5 (1.0–2.3)       | 0.020 (0.000–0.054) | 10                   |

## A.24 Matched group-size comparison

The three- to five-player pilots reported elsewhere in this supplement have no two-player arm, so a difference between group sizes could not be told apart from a difference between two separately built lanes. This subsection reports a matched comparison that removes that particular ambiguity.

Every group size, including two, is run through the same  $N$ -player engine, the same prompt template, the same parser and one protocol, the matched group-size protocol, version 1, so the number of seats is the only thing deliberately varied. The two-player arm is not a near neighbour of the headline game: evaluating the group-count stage rule of Equation 1 at two companies returns the two-player payoff matrix exactly, at 1.0, 0.6, 2.4, 2.0, which the released design check asserts offline before the task is pushed, together with the invariance of every other mechanism field and the sharing of the horizon draw described below.

*What was collected, and how.* The full grid: every group size at every private-risk condition,  $p_r^{\text{max}} \in \{0.1, 0.6, 0.9\}$ , ten repetitions each, on the admitted route google/gemini-3-flash-preview. That is 120 races and 3,696 decisions with zero parse failures. No collection account available to us sustains the whole four-by-three grid, which needs roughly 7,800 requests, so the sweep was partitioned into the analysis unit itself, one (group size, risk) cell of

ten repetitions, and each cell was collected whole on one identity so that it carries its own race-clustered interval rather than being stitched together across accounts. Cells were assigned to identities in a plan committed before the first was started, and the assignment rotates across risk levels so that no identity is confounded with a group size; every cell records its collecting identity in its collection receipt. Every cell’s bootstrap generator is derived from a digest of that cell alone, so collecting one condition cannot move an interval already reported for another, and the released analysis code refuses to report any risk level that is missing an arm.

Each cell was collected on a different account, again by a plan fixed in advance rather than by retrying until something succeeded:  $N = 2$  on one account,  $N = 3$ ,  $N = 4$  and  $N = 5$  on three others, each cell collected whole so that it carries its own race-clustered interval. The identity is a billing boundary rather than a model boundary, since the route, prompt, parser, protocol, temperature request and seed structure are identical across cells, but it is a difference between runs and is recorded per cell in its receipt rather than argued away.

*The contrast is paired on the horizon.* The game seed is base + rep and does not depend on the treatment name or on the number of seats, so one repetition index reuses a single horizon stopping-draw stream at every group size. Differencing a group size against the two-player arm inside a repetition therefore removes the horizon

**Table 24: Participant-level sensitivity on the confirmatory baseline, which holds all nine audited routes at ten races per risk level and so at the same matched size of 20. Columns, statistics, rarefaction and the 95% resampling interval (RI), which is not a confidence interval, are exactly those of Table 23; only the source of the trajectories differs, and the main paper’s admission verdict is printed beside each route. This table is the participant-level sensitivity for the primary dyad-clustered  $q=0$  comparison. Every admitted route sits entirely below the human Hill  $q=1$  interval at all three risk levels. The two routes that do not, GPT-5.4 mini and GPT-5.4 nano, are both routes the gate refused.**

| Risk | Population                      | $n_{src}$ | Hill $q=0$ (95% RI) | Hill $q=1$ (95% RI) | Hamming (95% RI)    | $n_{cluster}$ |
|------|---------------------------------|-----------|---------------------|---------------------|---------------------|---------------|
| 0.1  | Human                           | 98        | 19.2 (15.0–20.0)    | 18.9 (13.2–20.0)    | 0.467 (0.407–0.504) | 98            |
|      | Gemini 3 Flash (admitted)       | 20        | 3.0 (1.0–3.0)       | 1.5 (1.0–2.3)       | 0.020 (0.000–0.054) | 10            |
|      | Claude Opus 5 (admitted)        | 20        | 1.0 (1.0–1.0)       | 1.0 (1.0–1.0)       | 0.000 (0.000–0.000) | 10            |
|      | GPT-5.4 (admitted)              | 20        | 5.0 (3.0–5.0)       | 4.0 (1.9–4.9)       | 0.171 (0.076–0.234) | 10            |
|      | GPT-5.5 (admitted)              | 20        | 11.0 (3.0–11.0)     | 6.3 (2.2–8.9)       | 0.145 (0.058–0.216) | 10            |
|      | Claude Sonnet 5 (admitted)      | 20        | 2.0 (2.0–2.0)       | 2.0 (2.0–2.0)       | 0.211 (0.211–0.211) | 10            |
|      | Gemini 3.1 Flash Lite (refused) | 20        | 3.0 (1.0–3.0)       | 1.5 (1.0–2.3)       | 0.020 (0.000–0.054) | 10            |
|      | GPT-5.4 mini (refused)          | 20        | 20.0 (10.0–16.0)    | 20.0 (8.3–15.2)     | 0.416 (0.330–0.458) | 10            |
|      | Gemini 3.5 Flash Lite (refused) | 20        | 10.0 (6.0–10.0)     | 10.0 (5.0–9.5)      | 0.282 (0.200–0.331) | 10            |
|      | GPT-5.4 nano (refused)          | 20        | 16.0 (7.0–14.0)     | 15.2 (5.8–13.2)     | 0.418 (0.294–0.446) | 10            |
| 0.6  | Human                           | 104       | 19.7 (16.0–20.0)    | 19.7 (14.4–20.0)    | 0.495 (0.456–0.514) | 104           |
|      | Gemini 3 Flash (admitted)       | 20        | 14.0 (6.0–13.0)     | 12.3 (4.9–11.7)     | 0.333 (0.251–0.363) | 10            |
|      | Claude Opus 5 (admitted)        | 20        | 1.0 (1.0–1.0)       | 1.0 (1.0–1.0)       | 0.000 (0.000–0.000) | 10            |
|      | GPT-5.4 (admitted)              | 20        | 11.0 (5.0–11.0)     | 8.4 (3.4–10.0)      | 0.257 (0.145–0.297) | 10            |
|      | GPT-5.5 (admitted)              | 20        | 13.0 (5.0–13.0)     | 10.7 (4.2–11.5)     | 0.326 (0.259–0.352) | 10            |
|      | Claude Sonnet 5 (admitted)      | 20        | 5.0 (4.0–5.0)       | 4.9 (3.0–5.0)       | 0.180 (0.132–0.205) | 10            |
|      | Gemini 3.1 Flash Lite (refused) | 20        | 7.0 (3.0–7.0)       | 5.8 (3.0–6.8)       | 0.186 (0.094–0.254) | 10            |
|      | GPT-5.4 mini (refused)          | 20        | 17.0 (8.0–15.0)     | 16.2 (6.6–14.1)     | 0.360 (0.251–0.426) | 10            |
|      | Gemini 3.5 Flash Lite (refused) | 20        | 13.0 (7.0–13.0)     | 11.5 (4.8–11.7)     | 0.380 (0.290–0.411) | 10            |
|      | GPT-5.4 nano (refused)          | 20        | 19.0 (9.0–16.0)     | 18.7 (7.3–15.2)     | 0.459 (0.358–0.481) | 10            |
| 0.9  | Human                           | 138       | 19.5 (16.0–20.0)    | 19.3 (14.8–20.0)    | 0.497 (0.461–0.516) | 138           |
|      | Gemini 3 Flash (admitted)       | 20        | 12.0 (5.0–12.0)     | 10.0 (4.0–10.7)     | 0.414 (0.349–0.419) | 10            |
|      | Claude Opus 5 (admitted)        | 20        | 1.0 (1.0–1.0)       | 1.0 (1.0–1.0)       | 0.000 (0.000–0.000) | 10            |
|      | GPT-5.4 (admitted)              | 20        | 14.0 (6.0–13.0)     | 12.5 (5.0–12.3)     | 0.414 (0.251–0.480) | 10            |
|      | GPT-5.5 (admitted)              | 20        | 13.0 (5.0–12.0)     | 10.7 (3.9–10.7)     | 0.269 (0.155–0.349) | 10            |
|      | Claude Sonnet 5 (admitted)      | 20        | 7.0 (5.0–7.0)       | 6.0 (3.2–6.8)       | 0.267 (0.152–0.331) | 10            |
|      | Gemini 3.1 Flash Lite (refused) | 20        | 7.0 (5.0–7.0)       | 6.4 (3.4–6.8)       | 0.305 (0.194–0.371) | 10            |
|      | GPT-5.4 mini (refused)          | 20        | 17.0 (7.0–15.0)     | 15.2 (5.5–14.1)     | 0.408 (0.258–0.477) | 10            |
|      | Gemini 3.5 Flash Lite (refused) | 20        | 13.0 (7.0–13.0)     | 12.3 (5.6–11.7)     | 0.403 (0.316–0.417) | 10            |
|      | GPT-5.4 nano (refused)          | 20        | 16.0 (8.0–14.0)     | 15.2 (6.3–13.2)     | 0.412 (0.267–0.465) | 10            |

**Table 25: Behavioural-archetype coverage: share of each population’s neutral-lane player-races assigned to each human-fit archetype. Human row is the reference clustering ( $n=341$ ); each LLM row projects that checkpoint’s own no-persona player-races ( $n=60$  to 120) into the same space. This table draws on the full nine-checkpoint cross-model pilot and therefore includes GPT-5.6 Luna and Terra. Those starred rows are retained only for exploratory archetype and classifier analyses and are not part of the nine-route confirmatory baseline.**

| Population            | Cautious | Aggr./recip. | Catch-up | Persister |
|-----------------------|----------|--------------|----------|-----------|
| Human (reference)     | 39.0%    | 49.9%        | 5.6%     | 5.6%      |
| GPT-5-nano            | 99.2%    | 0%           | 0%       | 0.8%      |
| GPT-5.4-nano          | 51.7%    | 33.3%        | 3.3%     | 11.7%     |
| Gemini-3-Flash        | 1.7%     | 78.3%        | 20.0%    | 0%        |
| Gemini-3.1-Flash-Lite | 0%       | 83.3%        | 16.7%    | 0%        |
| Gemini-3.5-Flash-Lite | 0%       | 76.7%        | 23.3%    | 0%        |
| GPT-5.6 Luna          | 3.3%     | 72.5%        | 24.2%    | 0%        |
| GPT-5.6 Terra         | 14.2%    | 59.2%        | 26.7%    | 0%        |
| Claude Opus 5         | 66.7%    | 33.3%        | 0%       | 0%        |
| Claude Sonnet 5       | 77.5%    | 20.8%        | 1.7%     | 0%        |

draw from the comparison. The analyser re-derives that pairing

from the recorded game seed instead of assuming it, and it holds for all ten blocks in every contrast.

*What it shows.* UNSAFE play rises monotonically with the number of competing companies at every private-risk condition that leaves room for it to rise, and all six of those contrasts exclude zero. Because the two-player arm is the headline game itself, this also says something about the rest of the paper: the two-player results are the low end of that curve, not a special case beside it.

Figure 13 draws that grid as its first panel, beside the two other manipulations this supplement ran on the same game: the crossed representation design of §A.7 and the seat asymmetry of the neutral baseline. The figure sits in this subsection rather than with the representation design for two reasons. Group size is the only one of the three factors that moves play a long way, so it is the panel the figure leads with; and by this point in the supplement the other two designs have already been described in full, whereas a reader meeting the figure at §A.7 would meet this grid many pages before the subsection that defines it. Read as a surface rather than as a table, the grid also makes plain something the rows do not. Adding companies does not carry play up a plane. It carries play into the ceiling, and the five hatched cells are where the measurement stops being a measurement: those cells contained no safe decision at all,



**Figure 13: Three manipulations on the same game, and only one of them moves play a long way. (a)** The matched group-size grid of Table 26 drawn as a surface, on google/gemini-3-flash-preview, ten races per cell and 3,696 decisions in all. Colour is the UNSAFE rate; the second line in each cell is the number of safe decisions the cell actually contained. Five of the twelve cells contained none, and those are hatched, because a cell with no safe decision left in it has no room to move and its distance from the cells below is a floor rather than an estimate. The column is also not group size alone: playing safe while everyone else does pays +1.0 at  $N = 2$  and -0.2 at  $N = 5$ . **(b)** The crossed representation design of §A.7 on both routes selected for the crossed representation study, drawn on a common thirty-point vertical span so that a slope in the upper panel is the same slope as in the lower one. The lines run close to parallel and the interaction spans zero on both routes, -1.4 pp [-9.2, +3.5] on Claude Sonnet 5 and +5.7 pp [-2.0, +14.3] on Gemini 3 Flash, which is an imprecise estimate consistent with no interaction rather than a demonstration that there is none. Near-parallel is not flat, and both main effects are present on both routes: the race story lowers UNSAFE play by 3.4 and 10.8 points against the neutral frame, and the mapping under which Q means SAFE raises it by 8.3 and 9.5 points. The two contrasts are signed in opposite physical directions, which the figure now states in words; the code contrast is also an algebraic identity of the counterbalancing, exactly twice the route's excess use of the letter Q, so it says which response surface carries the effect and does not on its own explain it. The letter use is no fixed habit: Gemini 3 Flash emits Q on 77% of its turns when Q means UNSAFE and on 11% when it means SAFE. **(c)** The two seats, whose prompts are symmetric word for word, at the opening move of all 270 neutral-baseline races. Four routes open UNSAFE from both seats in all thirty of their races and sit on the marked ceiling, where a seat effect could not appear at all; one seat opened UNSAFE in none of its races and is marked at the floor. Pooled over all nine routes, with races resampled inside their own route, the opening gap is +9.3 pp [+5.9, +13.0], and over every round of the race it falls to +2.6 pp [+1.6, +3.7]. Three limits travel with the figure. Panel a is one route, so it does not separate a property of group size from a property of this route in groups. An interaction that spans zero is inconclusive rather than absent, the more so on the route whose interval reaches +14 points, so panel b is not evidence that representation is inert. And because four routes in panel c are at the ceiling in both seats and can only pull the average toward zero, its pooled gap is a floor on the seat asymmetry rather than a measure of its size. No interval here is corrected for asking three questions on one figure.

so their contrast against the two-player arm is bounded by the boundary and not by the route.

The size of the effect is what makes the grid worth having. Going from two companies to three adds 11.1 points at  $p_r^{\max} = 0.6$  and 11.0 at 0.9; going to four adds 28.2 and 28.1. The two risk conditions

**Table 26: Matched group-size comparison on google/gemini-3-flash-preview. Ten races per cell, zero parse failures throughout. Rates carry 95% percentile bootstrap intervals over races; the contrast is the mean difference against the two-player arm within a repetition block, with its interval taken over the ten blocks. Horizon pairing was verified for 10 of 10 blocks in every row. Rates pool decisions within a cell. A contrast is the mean over matched repetitions of the within-repetition difference in race-level rates, so a contrast need not equal the difference of the printed rates. At  $p_r^{\max} = 0.1$  every cell is at the ceiling, so its intervals have zero width: that is an absence of variation in the sample, not precision. The five-company cell at  $p_r^{\max} = 0.6$  is at the ceiling too, so its contrast there is truncated, which leaves the high-risk condition as the only one that measures the five-company effect against no boundary.**

| N                                                         | Decisions | UNSAFE rate (95% CI) | minus N=2 (95% CI) |                  |
|-----------------------------------------------------------|-----------|----------------------|--------------------|------------------|
| <i>Low private risk, <math>p_r^{\max} = 0.1</math></i>    |           |                      |                    |                  |
| 2                                                         | 176       | 100.0% (100.0–100.0) | baseline           |                  |
| 3                                                         | 264       | 100.0% (100.0–100.0) | +0.0 pp            | (+0.0 to +0.0)   |
| 4                                                         | 352       | 100.0% (100.0–100.0) | +0.0 pp            | (+0.0 to +0.0)   |
| 5                                                         | 440       | 100.0% (100.0–100.0) | +0.0 pp            | (+0.0 to +0.0)   |
| <i>Medium private risk, <math>p_r^{\max} = 0.6</math></i> |           |                      |                    |                  |
| 2                                                         | 176       | 68.8% (65.2–72.4)    | baseline           |                  |
| 3                                                         | 264       | 80.3% (76.0–83.8)    | +11.1 pp           | (+6.3 to +15.1)  |
| 4                                                         | 352       | 98.3% (95.6–100.0)   | +28.2 pp           | (+22.6 to +33.2) |
| 5                                                         | 440       | 100.0% (100.0–100.0) | +31.0 pp           | (+26.4 to +35.3) |
| <i>High private risk, <math>p_r^{\max} = 0.9</math></i>   |           |                      |                    |                  |
| 2                                                         | 176       | 58.0% (50.7–63.7)    | baseline           |                  |
| 3                                                         | 264       | 68.9% (62.3–74.7)    | +11.0 pp           | (+5.9 to +16.4)  |
| 4                                                         | 352       | 83.8% (80.6–86.2)    | +28.1 pp           | (+24.5 to +32.0) |
| 5                                                         | 440       | 96.1% (92.3–98.6)    | +40.0 pp           | (+33.2 to +47.4) |

were collected on different days and different identities, and they agree to a tenth of a point. That agreement holds for the paired contrast. The differences of the printed rates at four companies are 29.5 and 25.9 points. The five-company contrast is the exception, +31.0 against +40.0, and the reason is visible in the table rather than interesting: at  $p_r^{\max} = 0.6$  the five-company cell is already at 100%, so its contrast is truncated by the boundary and understates the effect. The high-risk condition is the only one that measures the five-company step against no ceiling.

The low risk condition is where the comparison becomes informative about its own limits. There every cell is at 100%, so the group-size contrast is exactly zero at every size. The right reading is not that group size stops mattering when private risk is low. It is that this route is already at the ceiling before any competitor is added, so the design has no room left in which an effect could appear. The same route reaches 98.9% and 100.0% UNSAFE at  $p_r^{\max} = 0.1$  in the two independent neutral-baseline runs of §A.5, collected under a different protocol on a different engine, so the ceiling is a property of the route at low risk rather than an artifact of this sweep. Read together with the ceiling at  $p_r^{\max} = 0.6$  for five companies, the pattern is that this measurement has power only where play is off

the boundary, which is a statement about the instrument and not about where the mechanism operates.

*What it does not show.* Group size cannot be separated from everything that comes with it, and no analysis repairs that. The group-count rule makes the stage payoff depend on how many companies there are, and the state description names each of them, so a larger group is also a different payoff and a longer prompt. The estimand is therefore the effect of group size *within this game’s own definition*. The comparison covers one route and ten races per cell, so it is not a claim about endpoints or about a population of models. It is regenerated by the released analysis code, which refuses to report a risk level unless all four group sizes are present, refuses a cell with a parse failure or the wrong seat count, and records the collecting identity of every cell it uses.

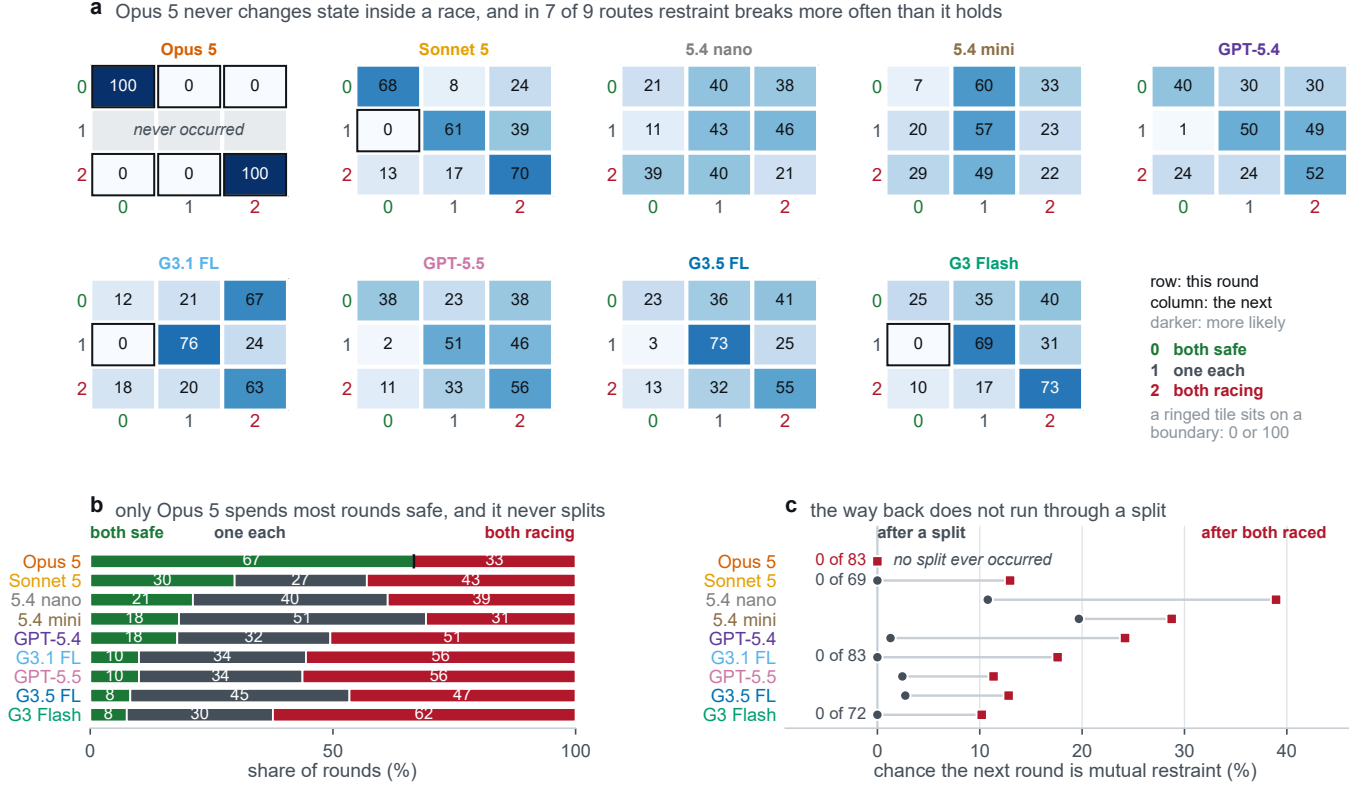
*Which route carries the seat gap.* Panel c of Figure 13 reports a gap averaged over nine routes, and a reader is entitled to ask whether it describes nine routes or one. The released analysis code answers that from the same corpus. The figures below weight each race equally, as the figure does. Over every round of the race the pooled gap of +2.6 pp is carried almost entirely by GPT-5.4 nano, at +19.5 pp [+12.9, +25.7]. Set that one route aside and the pooled gap falls to +0.5 pp [−0.4, +1.4]. Over the five admitted routes it is −0.5 pp [−1.3, +0.4], and Claude Opus 5 and Claude Sonnet 5 are at exactly 0.000 in all thirty of their races each. Weighting decisions instead of races gives +19.4 pp [+12.1, +25.9] for GPT-5.4 nano and a pooled gap of +2.3 pp. These intervals resample races without stratifying by route, so the pooled interval here is a little wider than the figure’s, which resamples inside each route; the point estimates agree exactly. The scripted-rival campaign agrees independently. There the rival is fixed and the route’s seat is counterbalanced five and five inside every cell, and the gap over its 60 cells, pooled across the five routes it covers, is −0.2 pp [−1.9, +1.6].

What a seat gap on a refused route says about the screen is a question for the concurrent submission [3], and this supplement does not argue it. The design protection this paper relies on is the seat counterbalancing. The usual bounds hold. This is nine endpoints in one game at one prompt version, nothing is randomised except the chair, and a route whose measured gap is zero is at zero because its two seats agree, not because the design removed the effect.

## A.25 Where a race goes, once it gets somewhere

Every behavioural number reported elsewhere in this paper describes one seat: it says what a route plays, given the risk it was told about or given what it saw its rival do. A race is not one seat. What happens to the world in a round is the *joint* outcome, and with two seats and two actions that outcome is one of three states: both safe, one each, and both racing. A race is then a walk on three states, which is the object the evolutionary lane of this paper reasons about and the object the behavioural lane did not have. Figure 14 builds that walk from the same nine-route neutral corpus used throughout: 270 races, 2,511 rounds, and 2,241 transitions between consecutive rounds of one race.

The two mixed outcomes are collapsed into a single *split* state. These are self-play races, so the two seats are the same route under



**Figure 14: The race as a walk on joint outcome states.** The joint outcome of a round is how many of the two seats played UNSAFE, giving three states: both safe, one each, and both racing. Panel a gives one transition matrix per route, rows this round and columns the next, each row summing to 100, ordered from the route that spends most of its rounds in mutual restraint to the route that spends least. A ringed tile sits exactly on a boundary of 0 or 100 rather than near one, and the grey row for Claude Opus 5 marks a state that never occurred rather than a value that is missing. Panel b divides each route’s rounds between the three states. Panel c draws the two roads back to mutual restraint against each other: pooled over the eight routes that ever changed state inside a race, a split is repaired on 5.8% of the rounds that follow one and both seats stand down together on 18.3% of the rounds that follow mutual escalation, so four arrivals at restraint in five are simultaneous. The figure is not a mechanism: a transition count cannot say that a route noticed the state, and the same matrix would follow from two seats reacting only to their own accumulated private risk. Rows pool the three risk levels, which move these dynamics hard. Both seats of every race are the same route, so every number here is what a route does against a copy of itself.

the same prompt and differ only by which chair the protocol sat them in, which makes Company\_1 unsafe and Company\_2 unsafe the same event under a relabelling. The regenerating script tests that exchangeability rather than asserting it: Company\_2 is the racing seat in 53.5% of the 818 split rounds, close enough to a coin that a four-state chain would buy noise rather than structure. Transitions are counted only between consecutive rounds of the same race, and the last round of a race is dropped because it has no successor. Each race draws its own horizon once, before the first move, so censoring cannot depend on where the walk had got to.

Three readings survive the figure. First, restraint is not sticky. Pooled over the eight routes that ever changed state inside a race, mutual restraint is followed by mutual restraint on only 34.4% of occasions, 105 of 305, and in seven of the nine routes the chance that restraint holds into the next round is below one half. Second, only anthropic/claude-opus-5@default spends most of its rounds

in mutual restraint, 66.7%, and it is also the one route whose two seats never once played differently. Third, the road back matters: a split is repaired on 5.8% of the rounds that follow one, 42 of 723, while both seats stand down together on 18.3% of the rounds that follow mutual escalation, 176 of 964. Of the 218 arrivals at mutual restraint, 80.7% are both seats standing down at the same moment rather than one seat stepping down and the other joining it a round later.

The Opus 5 matrix deserves its own sentence, because a pooled matrix with every count on the diagonal invites the word *frozen*, and that word would invert this paper’s own reading of the route. The constancy is within a race and not across the design. At  $p_r^{\max} = 0.1$  all ten of its races sat in mutual escalation for every round; at 0.6 and at 0.9 all twenty sat in mutual restraint. The route therefore moves further with risk than any of the other eight, a full 100 points of UNSAFE rate against 57 for the next most responsive. It is a step

in self-play: deterministic inside a cell, and the one route in the figure whose pooled matrix is an artifact of the pooling rather than a summary of it.

*What it does not show.* A transition count is not a mechanism. Nothing here says a route noticed the state, and the same matrix would be produced by two seats reacting only to their own accumulated private risk. Rows pool the three risk levels, and risk moves these dynamics hard, so a route’s matrix summarises three regimes rather than stating a law. The ordering in panel c is a property of the pooled rows and not of every regime inside them: at  $p_r^{\max} = 0.1$ , where mutual restraint is reached at all in only 4% of rounds, the two roads swap and repair from a split becomes the likelier one, 7.1% against 3.8%. Thirty races per route is enough to separate a route that never moves inside a race from a mobile one and not enough to pin a single cell to a point. And a state is only a joint action, so two routes can walk the same three states with entirely different progress, payoff and accumulated private risk behind them.

## A.26 The designed boundary at round five

The mechanism has one sharp, designed, disclosed edge in it, and it sits between round four and round five. The prompt every player reads says that the race lasts at least five rounds and that from round five onward each completed round ends the race with probability 20%. The continuation probability is therefore exactly 1 while the race is short of round five and 0.8 from round five onward. The rule is disclosed; the realised length is not. This is the one quantity in this game that the folk-theorem tradition says should move cooperative play, because a threat certain to be carried out sustains cooperation that a threat with a one-in-five chance of never arriving does not. Figure 15 asks whether the routes’ play moves there. The regenerating script parses the horizon sentence out of every prompt in the corpus rather than trusting the protocol document, so it fails rather than drawing a boundary the players were never told about.

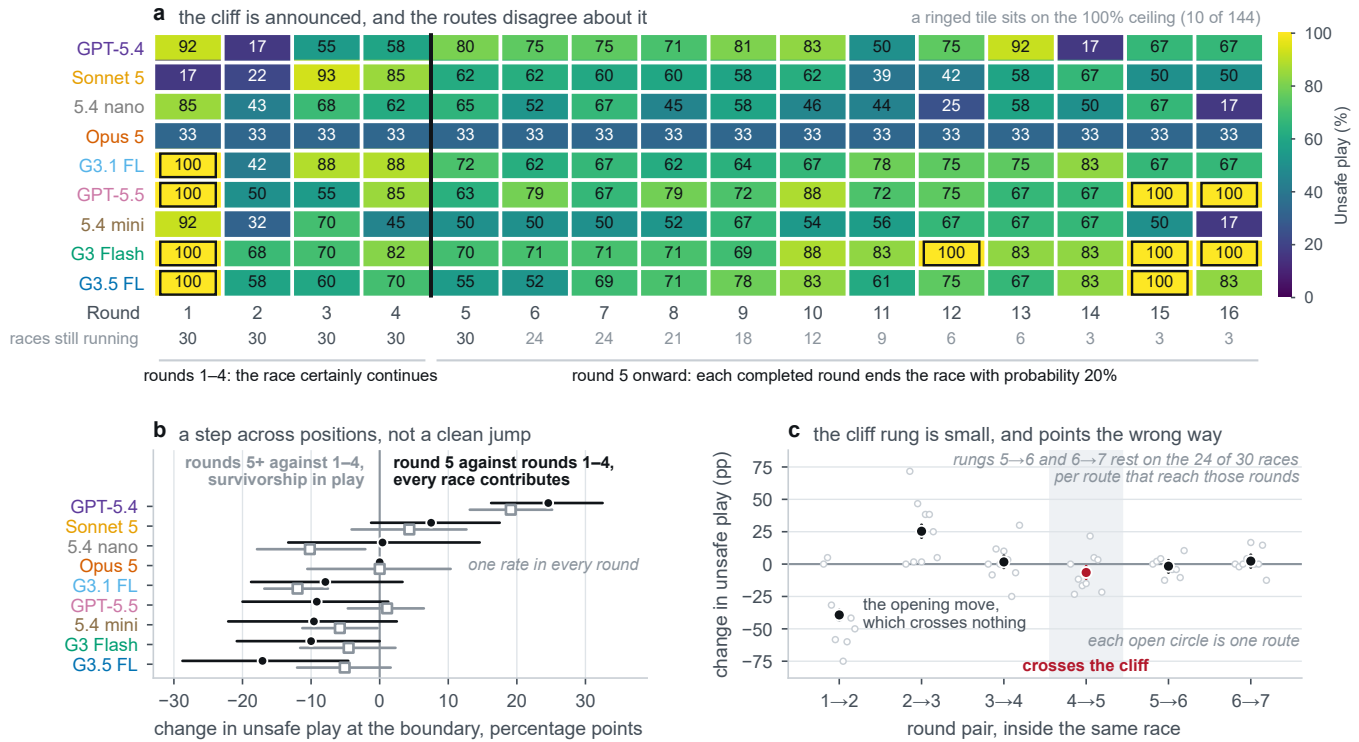
One route steps up at the boundary and one steps down. Comparing round five against rounds one to four, the window in which all thirty races per route contribute by construction, `openai/gpt-5.4-2026-03-05` plays 24.6 points more UNSAFE, with a 95% percentile interval of 16.2 to 32.5, and `google/gemini-3.5-flash-lite` plays 17.1 points less, with an interval of -28.8 to -4.6. The other seven intervals cover zero, several of them wide enough to hold a twenty-point move in either direction. `anthropic/claude-opus-5@default` plays one rate in every round of the corpus, so its step is exactly zero and could not have been anything else.

*Why this is a comparison and not a discontinuity estimate.* Two confounds sit between the design and any causal reading, and the figure carries both on its face rather than in a footnote. The first is accumulation: a later round is also a round with more history behind it and with more accumulated private risk, and because those move together inside a race no contrast between “before round five” and “round five onward” can separate them from the horizon. That is why panel b is labelled as a step across round positions, and why panel c exists at all. Comparing each race with itself one round later holds accumulation nearly fixed: every rung of that ladder is one round of extra history, and exactly one rung crosses the boundary. The second is survivorship. Races have different sampled lengths,

so the races that reach round twelve are not the races that reached round three. Every route was run against the same ten horizon draws at each of the three risk levels, so no route is favoured, but the race set still changes along the axis. The filled estimate in panel b and the first four rungs of panel c stay inside rounds one to five, which every race reaches by construction; the last two rungs rest on the twenty-four races per route that get that far, and the panel says so.

The ladder is where the argument lands. The rung that crosses the boundary is the only rung after the opening two whose interval clears zero, so it is not nothing: -6.5 points, interval -11.1 to -1.9. But it is small beside the opening move, -39.3 points from round one to round two, and beside the +25.4 points from round two to round three, and it is negative, which is the direction opposite to the one a shortening shadow of the future is supposed to push. The grey context in that panel is not decoration: three of the nine route means point the other way, so the pooled interval is a statement these nine routes make together rather than one that each of them makes, and resampling routes rather than races widens it to cover zero.

*What it does not show.* This is not a regression discontinuity. The running variable is a round index that also carries history, the design has no observations arbitrarily close to the threshold, and nothing here is a causal estimate of the horizon. Nor is it a test of whether these routes can reason about a horizon at all; it tests whether their play moves when the stated horizon changes, in this game, at this prompt version. The absence of a step is evidence about the step and not proof that the prompt sentence went unread: panel a shows one route whose play is a constant, and a constant cannot move at a boundary. Both seats of every race are the same route, so what is measured is a horizon response against a copy of itself and not against a different opponent. Nine commercial endpoints are not a sample from a population of models.



**Figure 15: UNSAFE play by round position, with the designed boundary at round five.** Panel a gives one row per route and one column per round, with the surviving race count printed under each column and a ring on the tiles that sit on the 100% ceiling, where the contrast above is truncated rather than measured. Panel b gives the step at the boundary per route under two windows: round five against rounds one to four, where all thirty races per route contribute, and all later rounds against rounds one to four, where survivorship and unequal decision counts are in play. Seven of the nine intervals cover zero and the two that do not point in opposite directions, +24.6 points for GPT-5.4 and −17.1 for Gemini 3.5 Flash Lite. Panel c is the placebo ladder: within-race change from one round to the next, pooled over routes, with the nine route means as grey context. *This is a comparison across round positions and not a discontinuity estimate.* A later round also carries more accumulated history and, because private risk is the fraction of a player’s own completed actions that were UNSAFE, more accumulated private risk; round position and accumulated state are collinear inside a race, so no contrast between the two sides of the boundary can separate the shortening horizon from what a later round otherwise brings with it. The rung that crosses the boundary is −6.5 points beside an opening move worth 39.3, and its sign is the opposite of the one a shortening shadow of the future is supposed to push.

## References

- [1] Gati V Aher, Rosa I. Arriaga, and Adam Tauman Kalai. 2023. Using Large Language Models to Simulate Multiple Humans and Replicate Human Subject Studies. In *Proceedings of the 40th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 202)*, Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett (Eds.). PMLR, Honolulu, Hawaii, USA, 337–371. <https://proceedings.mlr.press/v202/aher23a.html>
- [2] Elif Akata, Lion Schulz, Julian Coda-Forno, Seong Joon Oh, Matthias Bethge, and Eric Schulz. 2025. Playing repeated games with large language models. *Nature Human Behaviour* 9, 7 (01 Jul 2025), 1380–1390. doi:10.1038/s41562-025-02172-y
- [3] Anonymous. 2026. Who Passes Depends on the Questions: Validity Evidence for the Comprehension Screens that Admit Language-Model Agents to Strategic Studies. Concurrent submission, under review.
- [4] R. Maria del Rio-Chanona, Marco Pangallo, and Cars Hommes. 2025. Can Generative AI Agents Behave Like Humans? Evidence from Laboratory Market Experiments. arXiv:2505.07457 [econ.GN] <https://arxiv.org/abs/2505.07457> Preprint.
- [5] Elias Fernández Domingos and The Anh Han. 2026. Falling Behind Drives Unsafe Development in an Idealised AI Race Experiment. arXiv:2607.26034 [cs.AI] <https://arxiv.org/abs/2607.26034>
- [6] Oliver Gürtler, Lennart Struth, and Max Thon. 2023. Competition and risk-taking. *European Economic Review* 160 (2023), 104592. doi:10.1016/j.eurocorev.2023.104592
- [7] The Anh Han, Luís Moniz Pereira, Francisco C. Santos, and Tom Lenaerts. 2020. To Regulate or Not: A Social Dynamics Analysis of an Idealised AI Race. *Journal of Artificial Intelligence Research* 69 (2020), 881–921. doi:10.1613/jair.1.12225
- [8] Nathan Herr, Fernando Acero, Roberta Raileanu, María Pérez-Ortiz, and Zhibin Li. 2024. Are Large Language Models Strategic Decision Makers? A Study of Performance and Bias in Two-Player Non-Zero-Sum Games. arXiv:2407.04467 [cs.AI] <https://arxiv.org/abs/2407.04467>
- [9] Paraskevas V. Lekeas and Giorgos Stamatopoulos. 2026. What Suppresses Nash Equilibrium Play in Large Language Models? Mechanistic Evidence and Causal Control. arXiv:2604.27167 [cs.AI] <https://arxiv.org/abs/2604.27167>
- [10] Petra Nieken and Dirk Sliwka. 2010. Risk-taking tournaments – Theory and experimental evidence. *Journal of Economic Psychology* 31, 3 (2010), 254–268. doi:10.1016/j.joep.2009.03.009
- [11] Serkan Ozbeklik and Janet Kiholm Smith. 2017. Risk taking in competition: Evidence from match play golf tournaments. *Journal of Corporate Finance* 44 (2017), 506–523. doi:10.1016/j.jcorpfin.2014.05.003
- [12] Kenneth Payne and Baptiste Alloui-Cros. 2025. Strategic Intelligence in Large Language Models: Evidence from evolutionary Game Theory. arXiv:2507.02618 [cs.AI] <https://arxiv.org/abs/2507.02618>
- [13] Pouya Pezeshkpour and Estevam Hruschka. 2024. Large Language Models Sensitivity to the Order of Options in Multiple-Choice Questions. In *Findings of the Association for Computational Linguistics: NAACL 2024*. Association for Computational Linguistics, Mexico City, Mexico, 2006–2017. doi:10.18653/v1/2024.findings-naacl.130
- [14] Isaac Robinson and John Burden. 2025. Framing the Game: How Context Shapes LLM Decision-Making. arXiv:2503.04840 [cs.CL] <https://arxiv.org/abs/2503.04840>
- [15] Abel Salinas and Fred Morstatter. 2024. The Butterfly Effect of Altering Prompts: How Small Changes and Jailbreaks Affect Large Language Model Performance. In *Findings of the Association for Computational Linguistics: ACL 2024*. Association for Computational Linguistics, Bangkok, Thailand, 4629–4651. doi:10.18653/v1/2024.findings-acl.275
- [16] Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2024. Quantifying Language Models’ Sensitivity to Spurious Features in Prompt Design or: How I Learned to Start Worrying About Prompt Formatting. In *International Conference on Learning Representations*, Vol. 2024. OpenReview.net, Vienna, Austria, 25055–25083. [https://proceedings.iclr.cc/paper\\_files/paper/2024/hash/6c0e99d736da621403018ca7b32b1a4d-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2024/hash/6c0e99d736da621403018ca7b32b1a4d-Abstract-Conference.html)
- [17] Sheng-Lun Wei, Cheng-Kuang Wu, Hen-Hsen Huang, and Hsin-Hsi Chen. 2024. Unveiling Selection Biases: Exploring Order and Token Sensitivity in Large Language Models. In *Findings of the Association for Computational Linguistics: ACL 2024*. Association for Computational Linguistics, Bangkok, Thailand, 5598–5621. doi:10.18653/v1/2024.findings-acl.333
- [18] Zhaofeng Wu, Linlu Qiu, Alexis Ross, Ekin Akyürek, Boyuan Chen, Bailin Wang, Najeun Kim, Jacob Andreas, and Yoon Kim. 2024. Reasoning or Reciting? Exploring the Capabilities and Limitations of Language Models Through Counterfactual Tasks. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, Mexico City, Mexico, 1819–1862. doi:10.18653/v1/2024.naacl-long.102
- [19] Mingqian Zheng, Jiaxin Pei, Lajanugen Logeswaran, Moontae Lee, and David Jurgens. 2024. When “A Helpful Assistant” Is Not Really Helpful: Personas in System Prompts Do Not Improve Performances of Large Language Models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*. Association for Computational Linguistics, Miami, Florida, USA, 15126–15154. doi:10.18653/v1/2024.findings-emnlp.888